Summary

Fairness. Sustainability. Geographical representation. These are three key pillars of the International Olympic Committee's Agenda 2020+5 for organising global sports competitions such as the Olympic Games. These are also important goals particularly emphasised by the IMMC in creating the Global Sports League (GSL). In addition, the IMMC also emphasises on *logistical feasibility* of the schedule as an important factor to be considered.

However, pursuing all of these key goals may be an issue as some of these goals have inherent trade-offs with one another that have to be balanced in the creation of the schedule, such as fairness versus travelling distance and player welfare or economic benefits versus environmental impact. Due to multiple competing constraints that need to be balanced, creating a schedule for such a sports league while balancing these factors is a complex task. Some existing scheduling methods may fail to adequately account for factors such as fairness, logistical feasibility, and scheduling. To address the problem of balancing trade-offs, we developed a scheduling model that integrates these competing priorities in the creation of the GSL schedule.

In the GSL, to first select the 20 participating teams, we created a model that calculates the performance of each club using an **Elo rating system**, which is standardised to account for varying continental league strengths. The top few teams from each continental league in terms of performance (Elo score) were thus selected for the GSL. We also tested the validity of the Elo rating model by comparing the rankings of the teams made by our standardised Elo rating model to other rankings made by renowned football websites.

The overall scheduling model consists of two models: a **grouping model** and an **arrangement model** to create groupings and matchups for the group stage. Our grouping model splits the 20 teams into 4 groups using a data-driven approach that minimises the **incompatibility score**, which is affected by the skill difference (Elo score difference) and travelling distance (distance between home stadiums) of the teams in each group. We also conducted sensitivity analysis using **variance-based sensitivity analysis**, or **Sobol's method**, to appropriately calibrate and balance the weightage of the incompatibility score equation.

The arrangement model effectively determines the order for matches and club movement within a group during the group stage to ensure an efficient and equitable schedule that balances all the goals of the IMMC for the GSL. We employed **simulated annealing** in the arrangement model to tackle the problem of scheduling matches in the group stage, which is an example of the **Travelling Tournament Problem**. This model generates the order of the matches within each group which optimises travelling distance so that teams spend less time and energy on travelling between GSL matches, ensuring fair rest periods for all teams.

The overall schedule includes a **double round-robin group stage** and a **final playoff stage** hosted at Wembley Stadium, which was chosen for its sustainability and economic benefits.

Expanding the league and schedule by adding more teams would result in multiple impacts on the model in terms of the IMMC's goals. Firstly, adding more teams would improve geographical representation and economic sustainability, as there would be increased fan engagement with more fans from more teams watching the league and supporting their clubs. However, the overall schedule would also be tighter as more matches have to be played to reasonably determine the winner, thus impacting player welfare due to less rest time and reducing environmental sustainability due to more travelling. Our model is also generalisable beyond football and can be adapted to other sports with minor modifications, accounting for differences in game length, upset probabilities, and tournament structures.

Ultimately, our approach provides a robust framework for organising global sports leagues that align with IMMC's core objectives of ensuring fairness, sustainability, geographical representation and logistical feasibility.

Contents

Letter to the IMMC **4**

Overall Schedule **5**

1 Introduction **6**
1.1 Problem Restatement . . . 6
1.2 Chosen Sport and Reasoning . . . 6
1.3 Global Assumptions and Justifications . . . 6

2 Requirement 1: Selecting the Sport and Teams **7**
2.1 Distribution of Teams . . . 7
2.2 Local Elo Score . . . 8
2.3 Consideration Factors in Scheduling . . . 9
2.4 Trade-offs of Consideration Factors . . . 10

3 Requirement 2: Initial Schedule Development **10**
3.1 Overall Scheduling . . . 10
3.2 Justification for Number of Games for Group Stage . . . 10
3.3 Fair Grouping Model . . . 11
3.3.1 Variables . . . 11
3.3.2 Standardised Elo Rating (R'_{club}) . . . 11
3.3.3 Travelling Distance (ΔD_{AB}) . . . 11
3.3.4 Incompatibility Score (I_{AB}) . . . 12
3.4 Model Refinement . . . 12
3.4.1 Sensitivity Analysis . . . 12
3.4.2 Varying values of k . . . 12
3.4.3 Final Incompatibility Score . . . 13
3.5 Model Validation . . . 13
3.6 Group Allocation . . . 14
3.7 Match Ordering Model . . . 14
3.8 Overall League Schedule . . . 15
3.8.1 Group Stage Details . . . 15
3.8.2 Playoff Details . . . 16
3.9 Strengths and Limitations of Entire Model . . . 17
3.9.1 Strengths . . . 17
3.9.2 Limitations . . . 18
3.10 Achievement of IMMC Goals . . . 19
3.10.1 Achievement of Fairness . . . 19
3.10.2 Achievement of Sustainability . . . 19
3.10.3 Achievement of Geographical Representation . . . 20
3.10.4 Logistical Feasibility Aspect . . . 20
3.10.5 Number of Games to Determine Winner . . . 20

4 Requirement 3: Expanding the Model to More Teams **21**
4.1 Selecting More Teams . . . 21
4.2 Impact of Adding Teams on the Schedule . . . 21
4.2.1 Impacts on Fairness . . . 21
4.2.2 Impacts on Sustainability . . . 22
4.2.3 Impacts on Geographical Representation . . . 22
4.3 Impact on Number of Games Played . . . 22
4.4 Changes to Key Constraints and Impacts . . . 22

5 Requirement 4: Generalising the Model **23**
5.1 Adjusting the Number of Games . . . 23
5.2 Modifying Team Strength Measurement . . . 23
5.3 Competitive Balance Considerations . . . 24
5.4 Modifying the Match Ordering Model . . . 24
5.5 Number of games to be played . . . 24

6 Conclusion **24**

7 References **25**

8 Appendix **27**
8.1 Appendix A (Results and Data) . . . 27
8.2 Appendix B (Code) . . . 40
8.2.1 Generating Incompability Matrix . . . 40
8.2.2 Create Groupings . . . 41
8.2.3 TTP with Simulated Annealing . . . 44
8.2.4 Sensitivity Analysis (Sobol's Method) . . . 53

Letter to The IMMC

Dear Directors,

We appreciate the opportunity to design a groundbreaking scheduling model for the Global Sports League (GSL) that achieves IMMC's vision to allow for equitable travel time, distance and balanced competitiveness of matchups in football (soccer). We also considered many other factors like economical benefits, sustainability and geographical representation.

To ensure true global inclusivity, we guaranteed proportional representation from each continent while prioritising underrepresented regions — giving clubs and fans worldwide a stake in the GSL's success.

We pioneered a data-driven standardised Elo rating model, adjusted for overall league difficulty, to objectively rank teams. After that, we did a model validation to prove the effectiveness of our model. This ensured that clubs in stronger leagues had their Elo increased while clubs in weaker leagues had their Elo reduced.

We have broken down the GSL league schedule into 2 big parts; the group stage and playoffs. For the group stage, we allocated the groups by fixing 4 group "leaders". These group "leaders" were the top 4 strongest teams based on the standardised Elo rating model. Then, to ensure competitive equity and minimal travelling distance, we used an Incompatibility Score model that forms the groups by minimising Elo difference and travelling distance. This gives every team an equal shot at victory without exhausting cross-continent flights. Within each of the groups, we drew parallels to the Travelling Tournament Problem (TTP). By leveraging advanced computational methods (Simulated Annealing), we optimised the schedule to minimise total distance travelled while guaranteeing teams adequate rest — a first for international leagues. Our analysis found that this method led to a decrease in total distance travelled during the group stage by approximately half.

The playoffs will be held at Wembley Stadium. This is because not only does Wembley Stadium have incredible high seating capacity it is also incredibly eco-friendly and is located in a prime spot to maximise revenue from ticket sales, hence we chose to host the playoff games there.

Our solution is also adaptable to other sports with varying team dynamics and rules. We have suggested how they could be adapted and how to overcome some of the limitations other sports might have.

Our solution prioritises:

1. **Fairness:** No team is disadvantaged by excessive travel or imbalanced matchups.
2. **Sustainability:** Reduced carbon footprint through optimised travel and green stadiums.
3. **Economic & Fan Benefits:** Prime-time scheduling and iconic venues drive attendance, broadcasting revenue, and global engagement.

Our model acts as a blueprint for the future of GSL and achieves IMMC's goals. Attached below is the graphics of the proposed schedule of the GSL.

Best Regards,

Team Mathematics Fanatics

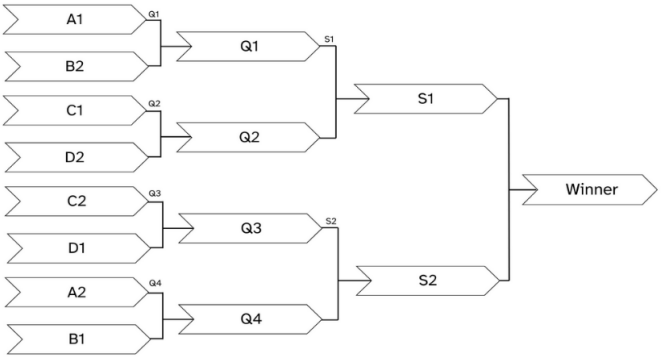
Overall Schedule

Group Stage

Month	Week	Group A	Group B	Group C	Group D
March	1	River Plate (H) vs Penarol	Manchester City vs Columbus Crew (H)	Atletico Mineiro vs Monterrey (H)	Al Ahly (H) vs ES Tunis
	2	Botafogo vs America (H)	Bayern Munchen (H) vs Borussia Dortmund	Auckland City vs A.S. Pirae (H)	Al-Hilal (H) vs Al Ain
	3	Pachuca (H) vs Penarol	Borussia Dortmund vs Yokohama F. Marinos (H)	Monterrey (H) vs Mamelodi Sundowns	Real Madrid vs ES Tunis (H)
	4	River Plate (H) vs America	Manchester City (H) vs Bayern Munchen	Atletico Mineiro (H) vs Auckland City	Al Ahly (H) vs Al Ain
April	1	Botafogo vs Pachuca (H)	Columbus Crew (H) vs Yokohama F. Marinos	Atletico Mineiro (H) vs A.S. Pirae	Al-Hilal vs Al Ahly (H)
	2	River Plate vs Penarol (H)	Manchester City vs Borussia Dortmund (H)	Monterrey (H) vs Auckland City	Real Madrid vs Al Ain (H)
	3	Pachuca vs America (H)	Manchester City vs Bayern Munchen (H)	Monterrey vs Mamelodi Sundowns (H)	Al-Hilal vs ES Tunis (H)
	4	River Plate vs Botafogo (H)	Borussia Dortmund (H) vs Columbus Crew	Auckland City (H) vs A.S. Pirae	Al Ahly vs Al Ain (H)
May	1	Penarol (H) vs America	Columbus Crew vs Yokohama F. Marinos (H)	Mamelodi Sundowns (H) vs A.S. Pirae	Real Madrid (H) vs Al Ain
	2	River Plate (H) vs Pachuca	Manchester City (H) vs Borussia Dortmund	Atletico Mineiro vs Auckland City (H)	Al Ahly vs ES Tunis (H)
	3	Botafogo (H) vs Penarol	Bayern Munchen vs Columbus Crew (H)	Monterrey vs A.S. Pirae (H)	Al-Hilal (H) vs ES Tunis
	4	Pachuca (H) vs America	Borussia Dortmund (H) vs Yokohama F. Marinos	Atletico Mineiro (H) vs Mamelodi Sundowns	Real Madrid vs Al Ahly (H)
June	1	River Plate (H) vs Botafogo	Manchester City (H) vs Columbus Crew	Mamelodi Sundowns (H) vs Auckland City	ES Tunis vs Al Ain (H)
	2	Penarol vs America (H)	Bayern Munchen (H) vs Yokohama F. Marinos	Atletico Mineiro vs A.S. Pirae (H)	Real Madrid vs Al-Hilal (H)
	3	River Plate vs Pachuca (H)	Manchester City (H) vs Yokohama F. Marinos	Monterrey vs Auckland City (H)	Real Madrid (H) vs ES Tunis
	4	Botafogo vs Penarol (H)	Bayern Munchen vs Borussia Dortmund (H)	Mamelodi Sundowns vs A.S. Pirae (H)	Al-Hilal (H) vs Al Ahly
July	1	River Plate vs America (H)	Bayern Munchen vs Yokohama F. Marinos (H)	Atletico Mineiro (H) vs Monterrey	ES Tunis (H) vs Al Ain
	2	Botafogo (H) vs Pachuca	Borussia Dortmund vs Columbus Crew (H)	Mamelodi Sundowns vs Auckland City (H)	Real Madrid (H) vs Al-Hilal
	3	Botafogo (H) vs America	Manchester City vs Yokohama F. Marinos (H)	Monterrey (H) vs A.S. Pirae	Al-Hilal vs Al Ain (H)
	4	Pachuca vs Penarol (H)	Bayern Munchen (H) vs Columbus Crew	Atletico Mineiro vs Mamelodi Sundowns (H)	Real Madrid (H) vs Al Ahly
Note: Top 2 from each group will proceed to the playoffs and each game will be held at Saturday 8pm local time					
August	**1 Month Break**				

Playoffs

Month	Week	Location	Playoff Stage	Match	
				Saturday, 8pm	Sunday, 2.30pm
September	1	Wembley Stadium	Quarter Finals	A1 v B2	C1 v D2
	2			C2 v D1	A2 v B1
	3			A1 v B2	C1 v D2
	4			C2 v D1	A2 v B2
October	1		Semi Finals	Q1 v Q2	Q3 v Q4
	2			Q1 v Q2	Q3 v Q4
	3	***Break***			
	4	Wembley Stadium	Finals	Saturday, 8pm	
				S1 v S2	



Legend:

A1, B1, C1, D1	First place team in the respective group
A2, B2, C2, D2	Second place team in the respective group
(H)	Location of match is the home stadium of that club

1 Introduction

There are many factors to take into account when it comes to scheduling. The problem is many of these factors have different trade-offs and maximising one objective could lead to sacrificing another. We hope to develop a scheduling model to minimise these trade offs and find a pareto optimal solution. IMMC prioritises fair distribution of games across teams, equitable travel time and distance, and balanced competitiveness of matchups. Therefore, we have developed a scheduling model that optimised these IMMC goals while also having additional key considerations: economic benefit, sustainability, scalability, player welfare, fan engagement and broadcasting effectiveness.

Our scheduling model splits the teams into groups and then optimises the order of games and timing of games within these to groups to need economical and environmental goals, along with other key aforementioned criteria.

1.1 Problem Restatement

1. Choose a team sport for the Global Sports League (GSL) and, using certain criteria, choose the 20 teams to participate. Additionally, consider the factors involved in creating a league schedule and their potential trade-offs.
2. Develop a model to create a competitively fair and logistically feasible schedule for the GSL.
3. Analyse the impact of adding 4 teams on our league schedule and also other key constraints on our model.
4. Explain ways to generalise our model for other team sports.
5. Communicate to non-experts our findings, suggested model, and schedule through a letter and graphic.

1.2 Chosen Sport and Reasoning

Football (soccer) was the chosen sport for the GSL. This was due to three major reasons:

1. Football is globally popular and the world's most popular sport as of 2025 [21]. With a large fan base, there are fans globally who will be attracted to watch the league and matches, thus the GSL would cater to the global audience.
2. Football also has well-developed leagues, with many national teams and clubs across continents. There is a top-level league in each continent for all the top clubs in that continent, such as UEFA Champions League, AFC Champions League and CAF Champions League. This means that many teams across the world can participate, which helps geographical representation.
3. There is also great infrastructure for football such as stadiums, which are developed in most countries, with each individual club also having their own stadium that serves as their own home stadium. These are locations that can be used for such league football matches, which helps logistical feasibility.

1.3 Global Assumptions and Justifications

1. The top few clubs in the highest-level league of each continent represent the top clubs of the entire continent.

   Justification: The reason that these clubs managed to reach the top of their continent's league is likely because they have good football performance, which would be the reason why we select these clubs from each continent.

2. Each country's club players do not play for other national teams.

   Justification: In order to standardise and compare the performance of individual clubs across the world, a method of obtaining the club's performance is required. The calculation of a club's Elo rating also depends on how well their country performs in general, as these leagues are only regional and do not consider the difference in skill between different regional leagues. Hence, the average Elo ranking of the countries were used to standardise each club's Elo across the world, using Elo data obtained from the official FIFA Elo rankings by country. Hence, it must be assumed that the country's club players do not play for other national teams in the FIFA World Cup so that the country Elo rankings are representative of the country's skill.

3. GSL is the only league the clubs are playing in.

   Justification: If there are any other leagues that the clubs are going to fly to other locations for, it will be difficult to get an accurate value for the total distance they are going to travel and thus the environmental impact of the event as it will be affected by other commitments that they might have and their locations.

4. Assume that travelling time and player fatigue (rest time) is not an issue because each team only plays a game at worst once a week, so they have ample time to travel between locations.

   Justification: Based on the double round-robin system, the group stage can be conducted over a period of 5 months, where 10 match slots can be allocated regularly with an interval of approximately 2 weeks. This would result in 10 matches per team if there are 24 teams in the league or 8 matches per team with 2 slots of rest breaks if there are 20 teams in the league.

2 Requirement 1: Selecting the Sport and Teams

2.1 Distribution of Teams

Variables	Remarks
T_t	Total teams participating in league
T_c	Total teams in the continent's top league
T_w	Total teams in top leagues for all 6 continents in the world

Table 1: Table of variables

The Global Sports League will be catered to football club teams. Before selecting the clubs that will participate, the distribution of the number of teams to each continent has to be calculated, by taking into consideration the prominence of the football scene in each continent, which ensures that each continent would have an equal chance to showcase the ability of their athletes.

This was done by calculating the number of clubs in the top league of the continent as a proportion of the number of clubs in the top leagues in the entire world. That fraction was then multiplied by the number of teams that would participate in the league, which is 20 teams, giving the formula below, which was used to determine what the number of teams from each continent would be:

$$N = \lfloor T_t \times \frac{T_c}{T_w} \rfloor \tag{1}$$

Continent	**Top League in Continent**	**Total teams in 2024 Season (T_c)**	**Teams representing in GSL (N)**	**Proportion of representation within each country**
North America	CONCACAF Champions Cup	30	4	0.13
South America	Copa Libertadores	32	4	0.125
Asia	AFC Champions League	24	3	0.125
Oceania	OFC Champions League	8	2	0.25
Africa	CAF Champions League	16	3	0.1875
Europe	UEFA Champions League	32	4	0.125
World Total (T_w)	-	142	20	-

Table 2: Number of teams representing each continent

However, by using the formula (1) alone, there would be a total of 18 teams selected globally, with 2 remaining empty slots that are unallocated. One of these team slots was allocated to Oceania, given their small size and underrepresentation in sports, with only New Zealand having attended the FIFA World Cup twice so far [8]. Furthermore, with the formula, they would only have 1 slot, which would not fulfil the 2 slots per continent criteria.

The other remaining slot was allocated to Africa, since the continent is also quite underrepresented in football geographically, with only 1 out of 11 teams making it to the World Cup in 2022 [3], the lowest of all the continents. Therefore, this additional slot would help increase the representation of Africa, helping to improve the competitive equity of the GSL.

2.2 Local Elo Score

Variable	Remarks
R_{club}	Elo of a club after the game
R_0	Elo of the club before the game, $R_0 = 800$ for all clubs at the start of the 23-24 season
R_{opp}	Elo of opponent club before the game
K	Scaling constant based on the importance of the match, matches with higher importance lead to more significant changes in Elo
W	Depends on the result of the game: Win - 1, Loss - 0, Draw - 0.5
W_e	Expected result of the match, calculated based on Elo difference between both teams

Table 3: Table of variables

The 20 teams from each continent were mainly selected based on the teams with the best performance in the six continental leagues in the 2023-24 season, as this would showcase the top team of each continent. The performance was calculated using an Elo rating system, which calculates the Elo score of a club based on their performance in the latest league season and their country's Elo from the Elo scores made by FIFA [7], the international governing body for football.

Data from the 2023-24 season for each club (Table 14-25, Appendix A) was used to calculate the club's Elo score within their continental league as it is the most representative of the current state of the skill level of the teams in that year. The formula used was similar to the formula that FIFA uses to calculate Elo scores [9], but with a small change, as the difference in Elo was divided by 400 instead of 600 to increase the spread of the Elo scores.

This increases the differences between the Elo scores of each club so that it can be used as a standard to group teams when matching them up. The following formulas were used to calculate the club's local Elo score within their continental league:

$$R_{club} = R_0 + K(W - W_e) \tag{2}$$

$$W_e = (10^{\frac{R_{opp} - R_0}{400}} + 1)^{-1} \tag{3}$$

Based on the Elo scores of all the clubs within each continent, these were the final 20 clubs that were selected for the league:

Continent	**Teams (Country)**
Asia (3 teams)	Al Hilal SFC (KSA), Yokohama F. Marinos (JPN), Al Ain FC (UAE)
Africa (3 teams)	Al Ahly FC (EGY), Espérance Sportive de Tunis (TUN), Mamelodi Sundowns F.C. (RSA)
North America (4 teams)	CF Monterrey (MEX), Columbus Crew (USA), Club América (MEX), CF Pachuca (MEX)
South America (4 teams)	Botafogo (BRA), Atlético Mineiro (BRA), River Plate (ARG), Peñarol (URY)
Oceania (2 teams)	Auckland City (NZL), A.S. Pirae (PYF)
Europe (4 teams)	Real Madrid C.F. (ESP), Borussia Dortmund (GER), FC Bayern München (GER), Manchester City (ENG)

Table 4: Teams chosen to represent each continent

2.3 Consideration Factors in Scheduling

1. **Distance:** A distance value calculated based on the locations of home stadiums of each club
2. **Balanced competitiveness of teams:** The difference between the standard of each team, which is determined by their Elo score in the Elo rating system
3. **Environmental sustainability:** The emissions generated due to the match. This includes emissions due to travelling, tourism, hosting the match itself etc.
4. **Economic sustainability:** The profit earned by the tournament organisers, as well as revenue gained by the host country/city
5. **Rest time between games:** Time elapsed since the last game played by the teams, be it within the league or outside the league
6. **Fan engagement:** The likelihood that fans would watch the match online and attend the match in the stadium
7. **Fair conditions:** Factors that may affect the performance of teams, such as playing many home or away games in a row.

2.4 Trade-offs of Consideration Factors

Prioritising the minimising of distance travelled may lead to sacrifices in the balanced competitiveness of the matches, as teams may be close to each other but have a high skill gap.

Environmental sustainability could also cause sacrifices in economic sustainability, as the environmental sustainability measures would increase costs and the stadiums which are the most environmentally friendly may not be the stadiums which would be able to attract a larger crowd.

Emphasising fair conditions, such as restricting the number of home or away games teams can play in a row, would also limit the scheduling options, which may increase the distance travelled by teams and thus negatively impact player welfare as well as environmental sustainability due to the increased emissions which are both caused by the increased amount of travelling done by players as well as necessary staff.

Balanced competitiveness may also impact economic sustainability, as stronger teams being matched in the same group as each other in the initial group stage would cause them to be eliminated before the playoff stage, which is the more anticipated stage for fans. Such balancing in the group stages could cause the playoffs to be less exciting and thus attract fewer fans.

3 Requirement 2: Initial Schedule Development

3.1 Overall Scheduling

The overall schedule plan is to have two separate stages of the league, the group stage and the playoff stage. The first stage, starting with all 20 teams, would be the group stage. This is where the 20 teams are split into 4 groups of 5 and play against each other, earning points based on their wins (3 points), draws (1 point) and losses (0 points). The top two teams of each group after playing against each other would move on to the playoff stage. The playoffs would consist of Quarter-finals and Semi-finals with a 2 leg system as well as finals with the winner being declared the winner of the league.

3.2 Justification for Number of Games for Group Stage

Firstly, the main purpose of a competition is to determine a winner that is the best team among the teams that participate in it. In sports, it is common for teams to undergo an upset, where a stronger team loses unexpectedly. Since such an upset may not be representative of the true standards of both teams, multiple games are required to reduce the likelihood of a team being eliminated due to upsets. To minimise this, an exponential function was used to determine the minimum number of games that need to be played between 2 teams to ensure that the winning team is likely to be the best team. This was done by taking reasonable doubt at 70% certainty that the winning team is the most capable team between a matchup of 2 teams.

As the number of games played approaches infinity, the certainty of the result (C) of the winning team truly being better approaches $C = 1$. Therefore, an exponential distribution was used to model the certainty of the result in the group stage, which has a cumulative distribution function with a formula $C = 1 - e^{-kn}$. Research shows that the probability of an upset is about 36% in football [2], however, this is estimated to be higher due to the presence of additional factors like differing weather conditions, illness, etc. Hence, the probability of upset was taken to be 45% and 0.55 was used as the constant k as it is the certainty that the team that wins is the more capable team. This gives the formula below for the games needed to be certain on the actual standards of the team:

$$C = 1 - e^{0.55n} \tag{4}$$

When $C = 0.70$, $n \approx 2$. Hence, it was decided for each team to play against one another at least twice during the group stage so that the result is a fair representation of the comparative strength within the entire group.

3.3 Fair Grouping Model

3.3.1 Variables

Variables	Remarks
R_{club}	Elo rating of club within the continent (Local Elo), unitless
R'_{club}	Elo rating of club after standardisation with the rest of the world, unitless
$R_{country}$	Elo rating of country based on country's Elo scores by FIFA, unitless
R_{world}	Average Elo score given by FIFA of all countries in the world, unitless
$S_{country}$	Strength of country, unitless
λ_{club}	Longitude of club's home stadium, in degrees (°)
Φ_{club}	Latitude of club's home stadium, in degrees (°)
ΔD_{AB}	Distance between stadium of club A and club B, in kilometres (km)
I_{AB}	Incompatibility value of matching club B with club A, unitless
r	Radius of Earth $\approx$ 6378 km

Table 5: Table of variables

3.3.2 Standardised Elo Rating (R'_{club})

For grouping, an incompatibility scoring model was used. This model takes into account the relative skill level of the clubs and the distance between the clubs. This makes it easier to minimise distance travelling and Elo difference to ensure fair games in group stages by minimising incompatibility.

The Elo ratings calculated in Section 2.2 is based on the club's performance within their own regional league. This value was thus standardised with the Elo scores of other clubs across the world. Since different leagues have different participating countries, the average skill level might vary for each league, thus standardisation of Elo ratings across the world is necessary.

This was done by first obtaining a fraction of the club country's Elo rating over the average Elo rating of all the countries in the world, which gives the country's strength as a fraction of the global average strength, which is $S_{country}$. The standardised Elo rating, R'_{club}, was then calculated by multiplying the local Elo rating with the country strength fraction (Table 10, Appendix A). This gives the equations for standardised Elo rating below:

$$S_{country} = \frac{R_{country}}{R_{world}} \tag{5}$$

$$R'_{club} = S_{country} \times R_{club} \tag{6}$$

3.3.3 Travelling Distance (ΔD_{AB})

Additionally, another factor to consider when grouping teams is to minimise the distance travelled by the teams. Therefore, the relative distance between the clubs is also considered. This ensures that the clubs are reasonably spaced, minimising travel distance, and time, improving player welfare and even reducing carbon emissions from travel, improving logistical feasibility and environmental sustainability.

The travelling distance was determined using the difference in distance between the home stadiums of the 2 teams that are playing against each other. The distance difference between the stadiums of the two teams, Team A and Team B, was calculated using the Haversine formula with the longitude

(λ_{club}) and latitude values (Φ_{club}) of the club stadiums (Table 10, Appendix A). The distance difference formula based on the Haversine formula [11] is given below.

$$\Delta D_{AB} = 2r \arcsin \sqrt{\sin^2 (\frac{\Phi_B - \Phi_A}{2}) + \cos(\Phi_A)\cos(\Phi_B)\sin^2(\frac{\lambda_B - \lambda_A}{2})} \tag{7}$$

3.3.4 Incompatibility Score (I_{AB})

In the final model, the groups are selected based on the distance from one another and the relative standardised Elo score difference. This is to ensure that the matches are fair and competitive equity is maintained. This is given by an incompatibility score, I_{AB}, between two teams Team A and Team B.

In order to improve competitive equity, large skill gaps should be minimised by minimising the difference in Elo rating between the two teams so that the teams that compete against one another have a similar likelihood of winning. To amplify the effect of large skill gaps, the Elo difference between teams is squared in the incompatibility score, ensuring that in cases with high Elo differences, the incompatibility score between these two teams is increased by a significant amount. This helps to prioritise fair matchups and discourage large skill disparities, which could otherwise lead to one-sided games and reduce the competitive equity of the tournament.

The square of the Elo difference is then added to the difference in distance between the two club stadiums (ΔD_{AB}) which is multiplied by k, a constant with units of km^{-1}, yet to be determined. The formula of the incompatibility score between two teams, Team A and Team B, is thus as shown below:

$$I_{AB} = (R'_A - R'_B)^2 + k\Delta D_{AB} \tag{8}$$

3.4 Model Refinement

3.4.1 Sensitivity Analysis

The model is meant to consider both Elo difference and distance difference as significant factors, but with Elo difference having a slightly larger impact on the model as competitive equity of the matches would be slightly more important than the travelling distance that the clubs would have to travel between locations. As a result, the value of k in the incompatibility score equation should be chosen such that the incompatibility score is slightly more sensitive to Elo difference than distance difference, which would appropriately balance the trade-off between competitive fairness (Elo difference) and logistical feasibility (travelling distance).

In order to conduct sensitivity analysis for the incompatibility score, variance-based sensitivity analysis, also known as the Sobol's method, was used. This method decomposes the output variance of the model into contributions of the individual parameters and their interactions and determines the contribution of each input parameter and their interactions to the overall model output variance [24]. A high output variance due to a factor would show that the model is more sensitive to that specific factor and vice versa.

3.4.2 Varying values of k

Sensitivity analysis was thus conducted for various integer values of k to determine a suitable value of k such that the equation's sensitivity to Elo difference and distance difference is appropriately balanced. By using Sobol's method on the incompatibility score equation using different values of k, a table of first order indices of the factors ($R'_A - R'_B$) and ΔD_{AB} was obtained as shown below (full table in Appendix A, Table 13)

Value of k	First-order Sobol Index of $R'_A - R'_B$	First-order Sobol Index of ΔD_{AB}
1	0.9855	0.0146
2	0.9440	0.0561
3	0.8822	0.1179
4	0.8080	0.1920
5	0.7292	0.2708

Table 6: Sensitivity Analysis Results using Sobol Method

Initially, the constant k was decided as 1 for simplicity. However, when conducting sensitivity analysis, it was found that 98.55% of the variance was driven solely by Elo difference, as opposed to the 1.46% driven by travelling distance. This means that if $k = 1$, the model would be extremely sensitive to Elo differences and very insensitive to the distance in difference, almost ignoring it entirely, which is not desirable. This would impact the overall groupings as distance would have too little impact on the incompatibility score, leading to impractical groupings and scheduling, such as frequent long flights from place to place for matches, which would affect the players and teams in the league, which is not desirable.

In the end, the value $k = 4$ was decided as it appropriately balanced the contribution of Elo difference and distance difference, with approximately 80% of variance attributed to Elo difference and 20% to travelling distance. This ensures that the groupings in the group stage is mostly determined by the Elo differences, prioritising competitive fairness, while ensuring that travelling distance still has a relevant but secondary impact on the incompatibility scores as well.

Overall, Elo difference would be the primary driver of incompatibility of the model, which is expected as the value of Elo difference is squared and is desired due to the overall priority on overall fairness. Additionally, with $k = 4$, the travel distance also contributes significantly (20%), giving it a meaningful weight and impact as a secondary factor in the grouping model. This version of the incompatibility score equation would allow for groups in the group stage to still be formed based on the location of the club's stadium. This improves logistical feasibility through practical schedules, and environmental sustainability through minimising carbon emissions from travelling. It also maintains overall competitive equity through the relatively high weightage of Elo differences while minimising travelling distance between the stadiums in the groups.

3.4.3 Final Incompatibility Score

Through sensitivity analysis, it was determined that the constant $k = 4$, resulting in the incompatibility score between two teams given below:

$$I_{AB} = (R'_A - R'_B)^2 + 4\Delta D_{AB} \tag{9}$$

3.5 Model Validation

To ensure that our Elo system is accurate, we conducted model validation by testing the standardised Elo scores against other club rankings gained from online sources. The ranking from these sources are that of May 2024, when the 2023-2024 football season had ended for most regions.

From Table 11 (Appendix A), while our standardised Elo rankings mostly follow the official rankings, differences in rankings are observed due to our standardised Elo model using the country strength fraction, which might overestimate the strength of a club that is from a country with a high Elorating. However, when comparing the ranking of the clubs within the continents, our Elo model is able to rank the clubs accurately based on comparisons with online rankings.

We realised that this could be due to problem that some players in the national teams may not play for clubs in their country, but instead play for other clubs, which causes our model that relies on the strength of the country as a multiplier to be inaccurate. We suggest that to improve on this, it is possible to add a scaling factor to the standardised Elo. We could scale it as such: $R'_{club} = h(S_{country})(R_{club})$, where h is the propensity of a club to attract top talent. This would better standardise the Elo of the clubs such they are more accurate when compared to official rankings.

3.6 Group Allocation

However, the problem with competitive equity arises, as if competitive equity is prioritised in the group stage, this could lead to the playoffs games being less fair as all the good teams would be placed in a single group. Since only 2 of the teams in this group would advance, there would be less popular and good teams that make it to the playoffs, potentially leading to disengagement from fans and even players on the court. To minimise this, the clubs with the 4 highest standardised Elo scores are split among the 4 groups by making them the group "leaders" of each group. This ensures that strong teams are spread out, and the playoffs would be fairer.

Based on the table of the Standardised Elo (Table 10, Appendix A), the group leaders for groups A, B, C, and D are River Plate, Manchester City, Atlético Mineiro, and Real Madrid respectively. The grouping model works by allocating clubs to a group based on the incompatibility score of each club with the leader of the group. The incompatibility scores of each club were calculated with respect to the leader, as shown in Table 12 (Appendix A).

Allocation of the additional 4 clubs in each group is done by minimising the total incompatibility score of all 4 groups, where the total incompatibility score of a single group is given by the sum of the incompatibility of each of the four group members with respect to their leader. A scatter plot can be used to represent the different combinations and their total incompatibility (Fig 1, Appendix A).

From the model, the minimum incompatibility score was calculated to be 1417071.34 when the groups are as shown below, and that arrangement of groups is as shown below:

Group A	**Group B**	**Group C**	**Group D**
Atlético Mineiro	River Plate	Botafogo	Peñarol
Al-Hilal	Al Ahly	ES Tunis	Monterrey
Al Ain	Manchester City	Mamelodi Sundowns	Columbus Crew
Yokohama F. Marinos	Bayern Munchen	Pachuca	Auckland City
Real Madrid	Borussia Dortmund	América	Pirae

Table 7: Groups formed for the group stage

3.7 Match Ordering Model

The goal of the match ordering model is to find a schedule with the minimum total distance travelled (cost), with no more than 3 consecutive home or away games allowed for any team (atmost), which is the lowest number allowed while still being solvable [6]. Another constraint is that no 2 teams cannot play each other in a row (norepeat). This is known as the Travelling Tournament Problem (TTP) [6].

A double round-robin format of 6 teams, with the 5 clubs in a group and a "dummy" team, is being used. This is because matches must be made of a pair of teams, thus models to allocate matches require an even number of teams per group, resulting in the need for the "dummy" team. Clubs challenging a "dummy" team would be able to rest during this allocation instead in the finalised schedule. The "dummy" team will not contribute to the total distance travelled.

We will be using simulated annealing (SA) [1] to solve this problem. SA essentially changes the schedule, searching around a neighbourhood of size $O(n^3)$, where n is the number of teams. Some of the configurations generated may not meet the norepeat and atmost criteria, which would incur a penalty for this schedule configuration.

The neighbourhood of a schedule is the set of schedules that can be obtained by applying one of 5 types of moves as below:

1. Swap the home/away roles of two teams, such that if team T_i plays home against team T_j in round k, and away against T_j in round l, when swapped, team T_i plays away against team T_j in round k, and home against T_j in round l.
2. Swap the matches happening in two particular rounds for all teams
3. Swap the schedules of two teams, except for when they play against each other.
4. Swap the games of one team at 2 particular rounds, and then update the schedule to produce a double round-robin tournament by changing the games played for necessary teams.
5. Swaps the games of two teams in a particular round, and then updates the schedule to produce a double round-robin tournament by changing the games played for necessary teams.

SA will randomly select a schedule and check to see the change in the objective function $C(S)$, ΔC. If $\Delta C < 0$, the move is applied. Otherwise, there is a $e^{-\frac{\Delta C}{T}}$ probability of the move being applied, with T being the temperature constant.

$$\text{The objective function is defined as } C(S) = \begin{cases} cost(S) \text{ if } S \text{ is feasible} \\ \sqrt{cost(S)^2 + [w \cdot f(nbv(S))]^2} \text{ otherwise} \end{cases} \tag{10}$$

where $nbv(s)$ is the number of violations of the norepeat and atmost constraints, w is the weight of the violation, and $f(x)$ is $1 + \sqrt{x}\ln(\frac{x}{2})$. This is to allow exploration of infeasible solutions, while still appropriately penalising them.

SA uses a variable counter that increases for every non-improving move implemented, which is reset to 0 when there is the best improving move found. The temperature is also updated to $T \times \beta$, where β is a fixed constant ¡ 1 when the counter hits an upper limit, maxC, and the counter is reset to 0. This is to decrease the probability of accepting a non-improving move over time, the temperature is decreased.

Every time a new solution is generated, the weight w is multiplied by θ if the solution is infeasible or divided by θ if the solution is feasible. This is used to balance the time spent exploring feasible and infeasible solutions.

SA will reheat by doubling the temperature at which the best solution was found when the number of times the counter hits the upper limit of maxP, which is the phase value. SA then terminates when the number of reheats reaches the limit of maxR.

After SA was done, matrices were produced for each individual group (Fig 2, Appendix A). Based on this, the group stage grouping was produced (as shown in the graphic). This helped to minimise the total distance travelled by about 50% (174283.0 to 88792 for group A).

3.8 Overall League Schedule

3.8.1 Group Stage Details

As shown in Table 7, the groups were decided by minimising the total incompatibility, therefore minimising distance travelled and ensuring fair games with low Elo difference, and groups consisting of 5 members each were formed.

For the group stage, since there would be a total of 20 games held per group, it was decided that the group stage should last 5 months, starting in March and concluding in July, which would mean that there would be one match held per week. The schedule for this was considered by using the output from the match ordering model (Table 7), and taking every round to be a period of 2 weeks. The matches were thus allocated between the 2 weeks by considering the amount of breaks it would give each team and balancing the amount of breaks evenly.

This match would be held every Saturday at 8 p.m., since this timing allows for a balance between stadium attendance as well as TV or online broadcasting. Saturday 8 p.m. is also advantageous since fans would not have work or school the next day, and it also falls into prime-time TV slots, thus maximising audience engagement.

Furthermore, this timing also competes less with other major sporting events, thus fans would be more able to attend the match. Thus, it allows us to maximise both stadium attendance and TV as well as streaming broadcasting audience, increasing the revenue earned and helping out the economic sustainability of the league [22].

The group stage would operate based on a points based system, where 3 points are awarded for a win, 1 for a draw, and 0 for a loss. This is so as to make games more exciting in the group stages [15], which would entice more fans to attend the matches, increasing revenue and thus helping the economic sustainability.

After the group stage, there would be a 1-month break for teams to prepare for the playoffs, perhaps by strategising [13] and recuperation [14], so that players can showcase their best performance during the playoffs, keeping the playoffs exciting.

3.8.2 Playoff Details

The playoffs will begin at the quarterfinals stage. The quarterfinals and semifinals stage would have 2 legs each, whereas the finals will only be 1 leg. The quarterfinals and semifinals are two-legged, in order to make the game more exciting [10].

For the playoff venue, multiple factors were considered, such as stadium capacity, emissions caused by hotel stays for spectators, electricity used in operations of the stadium, waste generated in the operations of the stadium, as well as the impact on ticket revenue.

Europe was chosen as it has the most popular established football leagues, as well as the highest number of fans [16]. Fans travelling to watch games within Europe will play a minimal role in carbon emissions compared to fans travelling globally. Hence, to minimise travel emissions we decided Europe to be the host continent.

Wembley Stadium was chosen since it is the 2nd largest stadium in Europe [18] and the hotels near it in London have one of the lowest average hotel stay emission factors in the world at 9.8 kg CO2 emitted per hotel night [20]. Furthermore, 100% of the electricity used in Wembley Stadium is generated from renewable sources [19], it has a 77% recycling rate for waste generated during events [23], and is a zero waste to landfill venue [5].

Due to its status as England's national football stadium, it would be able to draw in many fans due to its iconic status. Being located in London, fans would also easily be able to attend the matches due to the ease of transport and the high amount of public transportation available. Fans going to the stadium to watch the matches would also likely go for a tour of London and the UK, helping the economic sustainability of the event, as it would also bring about economic benefits to the host country.

The economic sustainability would also benefit due to the high seating capacity meaning more tickets can be sold, and the ease of transport as well as iconic status of the football stadium would increase the demand for a ticket. The sustainability efforts of Wembley Stadium and the low emissions

generated by the surrounding hotels make it such that the environmental impact of the event is lowered significantly, and the public transport availability also reduces the impact of fan transport to and from the stadium, thus the environmental sustainability would benefit.

The playoffs would take place over 2 months, from September to October, since that time period is found to be the best for revenue [22]. This is because October has fewer competing sports events, especially in Europe, thus more people would attend the match or watch it, as they would not be distracted by large sporting events. The weather in October is more favourable, and matches in October also have higher TV ratings historically, showing that the audience watching through TV or online would also be maximised.

The timing of the games every week would be Saturday 8 p.m. and Sunday 2.30 p.m. Sunday 2.30 p.m. was chosen since workers and students do not have to worry about staying up too late to watch the game due to having work or school the day after. It also prevents clashes with other major games that might occur later in the day, thus avoiding any conflicts that might leave fans unable to attend [22].

The two legs of the quarterfinals would take place over the 4 weeks of September, with 2 games per week at Saturday 8 p.m. and Sunday 2.30 p.m. The two legs of the semifinals then take place in the first 2 weeks of October, with the 2 games of the first leg in the first week and the remaining 2 games in the last week. After a 1-week break for the players to rest, the finals would occur on the Saturday of the last week of October at 8 p.m.

3.9 Strengths and Limitations of Entire Model

3.9.1 Strengths

The model integrates multiple quantifiable factors and categorises them into two key areas. First, the competitive performance of teams as determined by Elo ratings, and second, logistical considerations such as travel distance, ensuring a holistic approach to tournament structuring. By incorporating both skill-based and practical constraints, our model ensures fairness while maintaining feasibility.

Additionally, the model also takes into account the possibility of upsets affecting matches in the league. It handles this problem by introducing an exponential-based certainty function with a threshold to determine the optimal number of games needed. This reduces the risk of upsets, enabling the fair evaluation of teams. By setting a certainty threshold of 70%, this ensures that stronger teams have a fair opportunity to prove their superiority over weaker teams while maintaining competitive unpredictability.

For the grouping model, a strength is that the model also refines the local Elo ratings by adjusting for continental league strength, allowing for fair comparisons between teams from different regions. This adjustment ensures that teams from historically weaker leagues are not unfairly penalised by being unfairly matched against higher skilled clubs.

Another strength is also that the grouping model appropriately balances the incompatibility score equation. The use of Sobol's method ensures that the parameters within the model are well-calibrated, making it robust against minor variations in input data. It also meets the aim of the incompatibility score equation by mostly placing weightage on Elo difference to prioritise fairness, while also placing a substantial weightage on the travelling distance such that it serves a secondary factor instead of being negligible.

For the arrangement model, a strength is that using TTP would allow for lowered travel distances, which not only helps achieve the aim of equitable travel time and distance, but also helps the environmental and economic sustainability by being able to reduce the emissions caused by transit between matches and reduce transport costs.

TTP also prevents a team from gaining unfair advantages or disadvantages since limits that teams cannot play 3 home or away games in a row are set. A team would also not be playing against the same opponent consecutively, as that would not be a very interesting match and would not lead to high attendance and viewership from fans.

Lastly, SA is also a relatively simple approach to solve the TTP, compared to other methods such as mixed-integer linear programming which makes it a suitable technique to implement, given the limited timeframe to complete the report. SA is also a rather memory efficient method, as it only has to keep track of the current and best solution, as well as its costs and other parameters attached to it. This is highly suitable due to the lack of computational power available.

Overall, the schedule also prioritises environmental sustainability and uses eco-friendly means whenever possible. This ensures that the GSL can occur every year without significant negative environmental impacts.

3.9.2 Limitations

Despite its strengths, our scheduling model also has several limitations. One major constraint is the assumption of a fixed certainty threshold (e.g., 70%) in determining the number of group-stage matches. While this value is based on statistical reasoning, it may not be optimal for every tournament format or sport. In real-world applications, factors such as injury risks, home advantage, and tactical matchups could influence optimal scheduling beyond a simple probability threshold.

A limitation of our scheduling model is that the match ordering model code that is used to determine the order that the teams in the group stage play requires a lot of computational resources. Further increasing the number of teams in the league might cause there to not be enough memory in the computer used to process all the possible combinations of creating the groups in the group stage and determine the groupings that minimise the incompatibility score according to our model.

Additionally, our model heavily relies on Elo ratings, which, while effective, do not capture certain key dynamics such as recent form, tactical adaptability, or head-to-head performances. Given that team performances fluctuate due to external factors like coaching changes, player transfers, or injuries, relying solely on Elo scores may not fully reflect a team's actual strength at any given time.

Another limitation is that our model does not fully account for external scheduling constraints like venue availability, broadcasting slots, and national team commitments. We assume that since the teams own the stadium, it would be available, although it may not truly be the case. While our grouping strategy minimises disruptions, real-world tournament organisers may face additional constraints that require further modifications to the model.

On top of that, the choice of k-value in our incompatibility score equation, which determines the balance between travel and skill differences, was based on sensitivity analysis but remains somewhat arbitrary. A more data-driven approach that incorporates historical tournament data could help refine this parameter for greater accuracy.

By using TTP to allocate matches to teams, there is an extremely strong focus on distance, as well as environmental sustainability since it is affected by distance as distance is the deciding criteria for whether a solution is good, thus other things such as economic sustainability by scheduling around rivalries would be overlooked.

However, this could be fixed by adding some parameters to the code that would penalise the solution if some rivalries are not inside the solution, but it would require some time to code as well as calculate how much the penalty should be to not overly affect the distance optimisation.

Despite these limitations, our model provides a structured and mathematically sound framework for fair and efficient tournament planning, with potential for further refinement through real-world implementation and additional constraints.

3.10 Achievement of IMMC Goals

3.10.1 Achievement of Fairness

Firstly, the schedule achieves fairness by implementing a double round-robin format in the group stage, as most clubs will play the same number of matches, unless they advance to playoffs. The grouping model also minimises the differences in Elo ratings, or differences in standards, of the teams in each group, thus ensuring that those clubs in each group have relatively the same standard and there are no large skill gaps, ensuring fairness.

The grouping model also considers travel distance that a team would have to travel during the group stage. Minimising the distance that the teams in the group have to travel helps to improve fairness by minimising the travelling that the teams have to do, allowing them more time to rest and train between matches and not be affected by their travels.

In addition, the group stage also consists of home and away games, allowing teams to play against each other twice, once in their home stadium and once in the other team's home stadium. This helps to minimise home court advantage. For the playoffs, the stadium location, Wembley Stadium, was chosen as it is also not the home stadium of the selected participant clubs, which will prevent any team from gaining home field advantage over the other teams in the playoffs.

Although clubs in Oceania may have difficulty travelling, the matches are scheduled such that there are breaks in between games of between 1 and 5 weeks for the teams to travel to the next stadium, as well as rest and recover from the previous match. This helps to reduce the effect of the slightly longer travel distance for their group and thus improving fairness in the league.

3.10.2 Achievement of Sustainability

There are two aspects of sustainability that are prioritised by IMMC and addressed by the model, which are economic and environmental sustainability.

Firstly, the schedule achieves economic sustainability. This was first achieved through the grouping model which minimises the distance between the teams in each group at the group stage, reducing travel costs and thus the amount of money that is spent on travel, which can be used for other purposes, improving economic sustainability.

The round-robin home and away system used in the group stage also allows for more stadiums to be used, as opposed to using a central stadium, increasing economic sustainability as more countries would benefit from the economic impact of the boom in visitors. This is because these visitors would inject money into the local economy through their spending [12] [17].

This was also achieved through setting the times of the games to be Saturday 8pm and for the playoffs, Sunday 2.30pm as our research has found that there would be more fan attendance and generate more revenue. The stadium chosen for the playoffs, Wembley Stadium, was also chosen as it is the second-largest stadium in Europe, with high holding capacity for large numbers of spectators to support their favourite clubs during the playoffs. In addition, it is in Europe, which is one of the continents with the highest number of football fans [16]. This means that more football fans would be able and willing to travel to the stadium to watch the playoffs, thus improving economic revenue and sustainability through tourism and increased business.

Other than economic sustainability, the schedule also achieves environmental sustainability by firstly minimising the travelling distance between clubs that are grouped together. This reduces the amount of travelling that teams have to do between matches and thus reduces the carbon emissions due to travelling, such as flying by plane between countries. In addition, the chosen stadium for the playoffs, Wembley Stadium, is chosen as it is very environmentally friendly and is one of the greenest stadiums in the world [4], thus helping to improve the environmental sustainability of the league.

3.10.3 Achievement of Geographical Representation

Another IMMC goal that our schedule works towards is geographical representation. This was done by ensuring the proportion of clubs from each continent participating is approximately the same is approximately equal. Additionally, more teams were selected from continents that were found to be previously underrepresented in other football leagues and World Cups, further improving their representation in the league.

It also promotes football clubs from these continents by elevating them to the global level through the Global Sports League. By playing home and away games during the group stage, the matches are also held in different places globally, increasing geographical representation as fans and players of certain clubs can travel to these different countries and support their teams, experience new places and the culture of football in these places.

3.10.4 Logistical Feasibility Aspect

The selection of the coefficient k in the incompatibility score equation using sensitivity analysis also allows for groups in the group stage to still be formed based on the location of the club's stadium. This helps logistical feasibility through facilitating practical schedules as there is less travelling involved. The schedule also facilitates rest time as the matches for the group stage are held between 1 and 5 weeks apart, giving teams time to rest and recover from their previous match as well as travelling to the next match location and adapt to the environment there.

There is also a one-month break before the playoffs, which helps all teams to rest and recover from the group stage matches, as well as formulate strategies and tactics for the playoff matches. In addition, the schedule is logistically feasible as most of the matches are played in the home stadiums of participating clubs, which means that stadium location and booking will not be a big issue as the stadiums would be free for the clubs to use to play against the other teams in their group.

3.10.5 Number of Games to Determine Winner

Based on the number of games that will be played in the group stage and playoffs, there will be 13 games total that are needed to determine the winner for the two teams that make it into the finals. There will be 8 games for each team in the group stage. For the teams that make it into the playoffs, they will play a total of 10 games if they are eliminated in the quarter-finals, 12 games if they are eliminated in the semi-finals and 13 games if they make it to the finals. Hence, the teams will play between 8 and 13 games throughout the season. Each team has to reasonably play 13 games to be declared the winner at the conclusion of the season.

4 Requirement 3: Expanding the Model to More Teams

4.1 Selecting More Teams

Based on the formula (1), the new number of teams representing in the league was obtained, which are given in the table below:

Continent	Total teams (2024 Season) (T_c)	Number of teams representing in GSL (N)	Number of added teams	Proportion of Representation per Continent
North America	30	5	1	0.167
South America	32	5	1	0.15625
Asia	24	4	1	0.167
Oceania	8	2	0	0.25
Africa	16	3	0	0.1875
Europe	32	5	1	0.15625
World Total (T_w)	142	24	4	-

Table 8: New number of teams representing each continent

Using the formula would give the results shown in the table above, with the exception of Oceania having a value of 1 instead of 2 and Africa having a value of 2 instead of 3 based on the formula, similar to the initial case when there were 20 teams total in the league. However, by using the formula alone, there would be a total of 22 teams selected globally, with 2 remaining empty slots that are unallocated, which were thus allocated to Oceania and Africa again, for the same reasons previously mentioned in Section 2.1.

Based on the Elo model selection, the additional teams that were selected are:

Continent	**Club (Country)**
North America	UANL (MEX)
South America	Fluminense FC (BRA)
Asia	Pohang Steelers (KOR)
Europe	Atlético Madrid (ESP)

Table 9: Teams selected for extra slots

4.2 Impact of Adding Teams on the Schedule

4.2.1 Impacts on Fairness

With the addition of more teams, this would provide a wider range of the Elo of the teams, thus there would be more matches being played where the Elo and thus skill difference between the teams is minimal, increasing the number of fair matches and benefitting the fairness of the schedule.

The increased number of matches played would increase the travelling distance, as teams will have to travel to more matches, which would disproportionately affect teams located far from the rest of the teams, such as teams in Oceania, since they would have to travel further. Thus, this negatively affects the fairness of the schedule.

The greater matches played also reduces the rest time that each team can have, increasing the effect of the carry-over effect and teams would be more strongly affected by playing stronger teams before their current match, which we did not take into account heavily in our previous model as the rest time between matches was quite long.

4.2.2 Impacts on Sustainability

In terms of economic sustainability, with more teams participating in the league, that would cause there to be more fans and more stadiums at different locations that these fans would visit. This benefits economic sustainability as it allows for the economies of the new hosting countries to benefit, as well as the generation of more revenue for the league.

However, adding 4 more teams means that teams will have to play 2 more matches during the group stage. This means that there is more travelling that all teams will have to do, which increases travelling costs for the team. Besides that, for fans who want to support their club at the stadiums, they would also incur higher travelling costs of travelling to various countries to watch their club.

In terms of environmental sustainability, more teams in the league means that there is more travelling that the teams will have to do, which increases the impact of their travelling on the environment, through carbon emissions of teams, as well as fans, travelling from place to place. Besides that, more matches also means that there are more games that are being played in more stadiums, which increase the carbon emissions of operation of matches during the league.

4.2.3 Impacts on Geographical Representation

With more clubs from each continent participating in the league, this would mean that the proportion of clubs of each continent participating in the league would increase, improving geographical representation. The number of teams participating in the league and performing well from each continent would also likely increase, thus promoting them on a global level and allowing for their representation geographically. In addition, conducting the group stage in a double round-robin format with both home and away matches would allow more matches to be played in various continents and countries around the world, allowing for more geographical representation. Fans of some clubs can also travel to various other stadiums to watch and support their club.

4.3 Impact on Number of Games Played

With 20 teams, there are only 5 teams in each of the four groups in the group stage. This means that each team only has to play a total 8 games in the group stage, with an additional 2 games for every further stage the club progresses to (quarter-finals, semi-finals and finals), which gives a range of 8 to 13 total games that are played by a team and 13 games in total are played for the winner team to be declared. However, with 24 teams, there are now 6 teams in each group. This means that each team now has to play 10 games in the group stage due to an extra team in every group, which means that each team has to play 2 games with the added team in the group, one home and one away, which adds 2 games to the total games each club plays in the group stage. Therefore, the clubs play a range of 10 to 15 games and 15 games are played for the winner to be declared.

4.4 Changes to Key Constraints and Impacts

In terms of timing and schedule, adding 4 more teams to the league would also mean that the schedule is tighter as there are more games to manage. The number of games that need to be played in order to reasonably determine the league winner also increases.

In addition, it would also lead to scheduling issues. Firstly, in the schedule for 20 teams, there are 4 groups of 5 teams, thus in a 2-week match slot, there will always be a team, which is the team fighting the “dummy team”, that is resting while the 4 other teams are playing against each other. This gives two matches within the group in every 2-week slot. However, with 24 teams, there will be 4 groups of 6 teams, resulting in three matches within the group in every 2-week slot. This means that teams no longer have rest breaks and thus players will have less time to recover between each match during the group stage.

Other than that, adding 4 teams with different Elo ratings and home stadium locations will also affect the grouping model as it affects the various combinations of groupings that are available that would minimise Elo difference and travelling distance, thus impacting our model.

Since the travelling distance for all teams increases and the schedule of matches is tighter, it will also affect logistical feasibility overall.

5 Requirement 4: Generalising the Model

The existing model accounts for factors like Elo-based team strength, travel distance, rest time, and number games for an accurate skill level judgement. To adapt this model to different team sports, adjustments must be made to how these factors are accounted for, in order to reflect the differences in rules, game durations, team dynamics etc.

5.1 Adjusting the Number of Games

The number of games that need to be played in order to confidently decide on the better team would vary from sport to sport due to each game operating on a different set of rules. The probability of an upset can be adjusted in the model when determining the number of games to be played to reduce the impacts of an upset.

For example, for higher-scoring games like basketball, upsets might be less likely than in football. The certainty function $C = 1 - e^{-kn}$ might need adjustment, with a higher k value reflecting a lower upset probability. Games like volleyball have a lower scoring than basketball, but are still higher than that of football. The usage of a slightly higher k value than basketball but lower than that of football would be ideal.

For sports like baseball and cricket, since these sports involve multiple innings/overs, a single game can provide a clearer indication of a team's skill level, thus a single round-robin group stage may suffice in ensuring that teams that win are the best team.

5.2 Modifying Team Strength Measurement

Football is a low scoring game, hence a single goal is significant. However, other team sports with higher-scoring games such as basketball, a single goal is not so significant. As a result, an Elo system depending on wins or losses might not be appropriate for such games. In such cases, different factors depending on the sport, such as point differentials, can be incorporated into the Elo system to better reflect the different effects of strength difference of competing teams on competitive equity.

Some sports might also have different stages in the games, such as offensive and defensive stages, which teams might have different strengths in. Therefore, different Elo ratings can also be given to teams in these separate aspects based on the sport.

Unlike soccer, where a single goal is significant, games like basketball involves higher-scoring games, thus a point differential-based Elo system may be more appropriate than a win-loss-based system to reflect true team strength.

Sports like cricket and baseball rely on batting or pitching matchups, which depends more on individual players, thus individual player performance should be weighted higher in team strength calculations. Therefore, metrics reflecting the performance of individual players, like on-base percentage in baseball, or strike rate in cricket, could be integrated into Elo calculations.

For games that are fast-paced with a high-scoring nature like lacrosse, the momentum can change rapidly within the game. The variation in team strategies, offensive schemes, and defensive setups can lead to upsets, especially in games where the offensive team has a hot streak. The Elo system

could thus be adjusted to account for offensive and defensive performances. For example, the number of goals scored or allowed per game could be integrated into the calculation to better reflect a team's true strength.

5.3 Competitive Balance Considerations

In some sports, one-sided matchups are more extreme, especially when the sport is high scoring and the winning team is almost always the better team. As a result, the power to which the Elo difference is raised can be varied from 2 based on how extreme one-sided matchups are in the sport.

For games like American Football and Baseball, while blowouts are still possible, the weaker team can sometimes win through strategy and game plans, or even sheer luck. In such cases, it may be acceptable for the $(R'_A - R'_B)^2$ term to remain the same, or even change the power to which the term is raised to 1.5 instead of 2, as the Elo difference is less likely to affect the quality of a game.

However, for games like basketball, a weaker team is far less likely to win due to the higher number of scoring events. Thus, the power of the Elo difference in the incompatibility formula might need adjusting. Changing the power being raised to 3 instead of 2 in high-scoring sports can be used to exaggerate the effect of mismatches to avoid boring and unfair games.

5.4 Modifying the Match Ordering Model

For games which do not require a double round-robin system such as baseball and cricket due to the length and format of the games, the TTP algorithm could be modified such that it would consider how many home and away games a team has already played compared to other teams, such that there would be an equal number of home and away matches played per team. It should also be adjusted to give higher resting times between games.

There would also be a change in the number of moves that can be made, since the 1st move of SA would no longer be able to be implemented, which may affect the speed at which an optimal solution would be found.

5.5 Number of games to be played

Based on the number of games that need to be played by teams to determine the winner, the schedule could be tightened or relaxed based on the sport and number of teams, which would also affect the intervals between matches and thus the scheduling of matches by the model.

6 Conclusion

Overall, we developed an optimised scheduling model for the GSL, balancing multiple key factors such as fairness, travel distance, and competitive equity. Breaking the model into two parts: the grouping and the arrangement model, helps us to optimise the grouping of teams in the group stage as well as optimise the order they play matches in during the group stage. This was done by balancing team skill differences and travelling distances, enhancing the fairness of the schedule obtained from our scheduling model. Sensitivity analysis of our model also shows that the incompatibility model is appropriately balanced, focusing primarily on fair groupgins of the teams, while also taking into account the travelling distance that teams have to travel as a secondary factor. Therefore, our overall scheduling model provides a structured and adaptable framework for league organisers to create fair schedules that can balance all of the aims of the IMMC, namely fairness, geographical representation and sustainability. Ultimately, our model approach presents a logistically feasible solution for scheduling competitive tournaments that optimises the balance of the IMMC's goals, ensuring an optimal experience for players, teams, and fans alike.

7 References

References

[1] A. Anagnostopoulos et al. "A simulated annealing approach to the traveling tournament problem". In: *Journal of Scheduling* 9 (Apr. 2006), pp. 177–193. DOI: 10.1007/s10951-006-7187-8. (Visited on 01/15/2022).

[2] Raquel Y.S. Aoki, Renato M. Assuncao, and Pedro O.S. Vaz de Melo. "Luck is Hard to Beat". In: *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (Aug. 2017). DOI: 10.1145/3097983.3098045. (Visited on 11/28/2020).

[3] Martin Armstrong. *Infographic: Where a World Cup Spot Is Like Gold Dust*. Statista Daily Data, Nov. 2022. URL: https://www.statista.com/chart/28822/teams-at-qualifying-stage-playing-for-one-world-cup-berth-by-region (visited on 03/28/2025).

[4] The Football Association. *Wembley Stadium celebrates green goals*. www.wembleystadium.com, 2014. URL: https://www.wembleystadium.com/news/2023/05/24/12/54/Wembley-Stadium-celebrates-green-goals.

[5] Matthew Campelli. *Wembley Stadium recognised for sustainability management system – Sustainability Report*. The Sustainability Report, Sept. 2019. URL: https://sustainabilityreport.com/2019/09/09/wembley-stadium-recognised-for-sustainability-management-system/.

[6] Kelly Easton, George Nemhauser, and Michael Trick. *The Traveling Tournament Problem Description and Benchmarks*. Ed. by Toby Walsh. Springer Link, 2001. DOI: 10.1007/3-540-45578-7_43. URL: https://link.springer.com/chapter/10.1007%2F3-540-45578-7_43#citeas (visited on 11/08/2021).

[7] FIFA. *Men's Ranking*. inside.fifa.com, Dec. 2024. URL: https://inside.fifa.com/fifa-world-ranking/men.

[8] FIFA. *Overview of the Participating Member Associations*. June 2011. URL: https://digitalhub.fifa.com/m/52463a2cc01d135/original/c91tgtfetbsmneex1vbc-pdf.pdf (visited on 03/28/2025).

[9] FIFA. *Revision of the FIFA / Coca-Cola World Ranking Overview*. URL: https://digitalhub.fifa.com/m/f99da4f73212220/original/edbm045h0udbwkqew35a-pdf.pdf.

[10] Ramón Flores et al. "What is a good result in the first leg of a two-legged football match?" In: *European Journal of Operational Research* 247 (Dec. 2015), pp. 641–647. DOI: 10.1016/j.ejor.2015.05.076. (Visited on 03/14/2021).

[11] GeeksforGeeks. *Haversine formula to find distance between two points on a sphere*. GeeksforGeeks, Aug. 2018. URL: https://www.geeksforgeeks.org/haversine-formula-to-find-distance-between-two-points-on-a-sphere/.

[12] Sports Business Group. *Sports Business Group Economic Impact of the NFL on London and the UK*. Oct. 2014. URL: https://www.london.gov.uk/sites/default/files/8925_attachment.pdf.

[13] Aron Laxdal et al. "Are the playoffs different from the regular season? A comparison of in-game statistics in Icelandic elite handball". In: *International Journal of Performance Analysis in Sport* 22 (Sept. 2022), pp. 1–7. DOI: 10.1080/24748668.2022.2115781.

[14] André P. Marques et al. "Periods of Competitive Break in Soccer: Implications on Individual and Collective Performance". In: *The Open Sports Sciences Journal* 15 (Jan. 2022). DOI: 10.2174/1875399x-v15-e2112141.

[15] GIANCARLO MOSCHINI. "INCENTIVES AND OUTCOMES IN A STRATEGIC SETTING: THE 3-POINTS-FOR-A-WIN SYSTEM IN SOCCER". In: *Economic Inquiry* 48 (Jan. 2010), pp. 65–79. DOI: 10.1111/j.1465-7295.2008.00177.x. (Visited on 01/16/2021).

[16] Felix Richter. *Infographic: The Global Game of Football*. Statista Infographics, June 2018. URL: https://www.statista.com/chart/14329/global-interest-in-football/.

[17] Robert Sandy, Peter J. Sloane, and Mark S. Rosentraub. *The Economics of Sport*. Macmillan Education UK, 2004. DOI: 10.1007/978-0-230-37403-4.

[18] Wembley Stadium. *The Stadium Facts and Features*. nhc-ws-cd.ncms-ase1.p.azurewebsites.net, 2022. URL: https://www.wembleystadium.com/about/stadium-facts-and-features.

[19] Wembley Stadium. *Wembley Stadium*. nhc-ws-cd.ncms-ase1.p.azurewebsites.net. URL: https://www.wembleystadium.com/about/sustainability.

[20] Staze. *Hotel Pollution Index - Europe*. Staze.com, 2023. URL: https://staze.com/blog/hotel-pollution-index/europe (visited on 03/28/2025).

[21] Eleni Veroutsos. *The most popular sports in the world*. WorldAtlas, Sept. 2023. URL: https://www.worldatlas.com/articles/what-are-the-most-popular-sports-in-the-world.html (visited on 03/26/2025).

[22] Chang Wang, Dries Goossens, and Martina Vandebroek. "The Impact of the Soccer Schedule on TV Viewership and Stadium Attendance". In: *Journal of Sports Economics* 19 (Jan. 2016), pp. 82–112. DOI: 10.1177/1527002515612875.

[23] Future Waste. *Wembley Stadium and Veolia celebrate green goals - Future Waste*. Future Waste, May 2023. URL: https://futurewaste.co.uk/wembley-stadium-and-veolia-celebrate-green-goals/ (visited on 03/28/2025).

[24] X-Y Zhang et al. "Sobol Sensitivity Analysis: A Tool to Guide the Development and Evaluation of Systems Pharmacology Models". In: *CPT: Pharmacometrics Systems Pharmacology* 4 (Feb. 2015), pp. 69–79. DOI: 10.1002/psp4.6.

8 Appendix

8.1 Appendix A (Results and Data)

Team	Country	Current ELO	Country ELO	Nation Scale	Real Elo	Latitude	Longitude
River Plate	ARG	857.98	1855.2	1.5143	1299.3	-34.545	-58.449
Manchester City	ENG	863.69	1800.05	1.4693	1269.1	53.483	-2.2
Atlético Mineiro	BRA	864.11	1784.09	1.4563	1258.4	-19.929	-44.014
Real Madrid	ESP	884.98	1732.64	1.4143	1251.7	40.453	-3.6883
Botafogo	BRA	854.42	1784.09	1.4563	1244.3	-22.893	-43.292
Pachuca	MEX	850.3	1652.7	1.349	1147.1	20.105	-98.756
Bayern Munchen	GER	858.85	1631.22	1.3315	1143.6	48.218	11.624
Peñarol	URU	840.04	1665.99	1.3599	1142.4	-34.796	-56.067
Borussia Dortmund	GER	853.14	1631.22	1.3315	1136	51.492	7.4516
Columbus Crew	USA	832	1665.27	1.3593	1131	39.968	-83.017
América	MEX	832.49	1652.7	1.349	1123.1	19.303	-99.15
Monterrey	MEX	832.24	1652.7	1.349	1122.8	25.669	-100.24
Yokohama F. Marinos	JPN	845.11	1620.19	1.3225	1117.7	35.51	139.6
Al Ahly	EGY	857.56	1518.91	1.2398	1063.3	30.069	31.312
ES Tunis	TUN	833.34	1523.23	1.2434	1036.2	36.747	10.273
Al-Hilal	KSA	858.64	1421.06	1.16	996.02	24.788	46.839
Al Ain	UAE	847.17	1364.46	1.1138	943.58	24.245	55.716
Mamelodi Sundowns	RSA	822.03	1357.08	1.1077	910.62	-25.753	28.222
Auckland City	NZL	839.69	1197.47	0.97748	820.79	-36.897	174.73
Pirae	TAH	834.41	999.48	0.81586	680.77	-17.54	-149.55

Table 10: Club Elo Data

Ranking	Real Elo Ranking	Football Database Ranking	IFFB Ranking
1	River Plate	Manchester City	Manchester City
2	Manchester City	Real Madrid	Real Madrid
3	Atlético Mineiro	Borussia Dortmund	Bayern Munchen
4	Real Madrid	Bayern Munchen	Al Ahly
5	Botafogo	Atlético Mineiro	Borussia Dortmund
6	Pachuca	River Plate	Botafogo
7	Bayern Munchen	Al-Hilal	Atlético Mineiro
8	Peñarol	Al Ahly	River Plate
9	Borussia Dortmund	América	Al-Hilal
10	Columbus Crew	Botafogo	Monterrey
11	América	Peñarol	América

12	Monterrey	Mamelodi Sundowns	Yokohama F. Marinos
13	Yokohama F. Marinos	Pachuca	ES Tunis
14	Al Ahly	ES Tunis	Mamelodi Sundowns
15	ES Tunis	Monterrey	Al Ain
16	Al-Hilal	Auckland City	Pachuca
17	Al Ain	Al Ain	Peñarol
18	Mamelodi Sundowns	Yokohama F. Marinos	Columbus Crew
19	Auckland City	Columbus Crew	Auckland City
20	Pirae	Pirae	Pirae

Table 11: Comparison of Elo Rankings

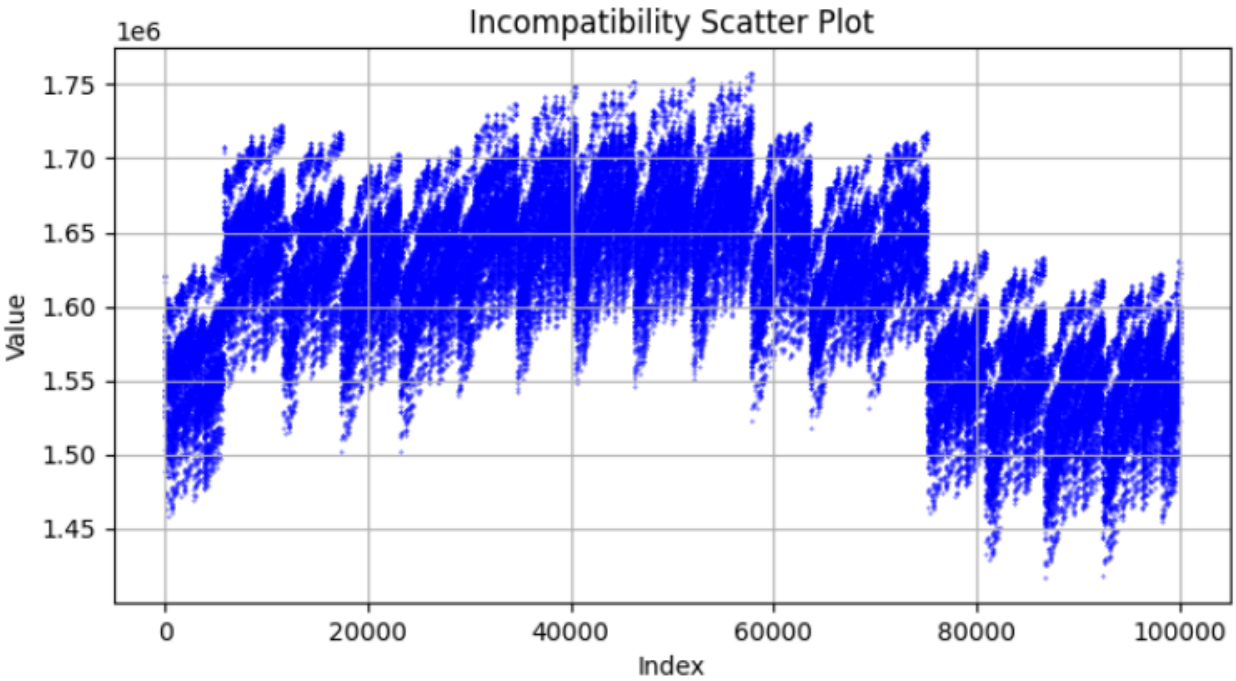


Figure 1: Scatter Plot of Incompatibility Configurations

Group	**A**	**B**	**C**	**D**
Group Leaders	River Plate	Manchester City	Atlético Mineiro	Real Madrid
Clubs	**Incompatibility Score Referenced to Leaders**			
Botafogo	10868.3	38077.39	1550.66	32647.29
Pachuca	52843.99	49445.94	42142.43	46865.13
Bayern Munchen	70342.3	20265.98	50681	17631.15
Peñarol	25500.39	60443.76	21597.62	51683.87
Borussia Dortmund	72729.13	20466.94	52521.9	19360.69
Columbus Crew	62979.07	43050.5	47409.84	40516.89
América	60540.97	56267.64	48011.67	52821.53
Monterrey	63225.2	54389.34	50046.72	51442.51
Yokohama F. Marinos	106462.86	60748.31	92794.46	61035.2
Al Ahly	103010.99	57342.88	77222.07	48913.09
ES Tunis	111818.03	62625.66	83364.96	51553.36
Al-Hilal	143482	95144.68	112873.99	85219.51
Al Ain	181125.52	128997.14	146529.14	117919.72

Mamelodi Sundowns	183626.56	165635.12	150338.89	148577.56
Auckland City	270402.85	273590.2	241549.81	264044.09
Pirae	418642.05	406621.48	377244.86	389418.02

Table 12: Incompatability score for Groupings

Value of k	**First-order Sobol Index of $(R'_A - R'_B)$**	**First-order Sobol Index of ΔD_{AB}**
1	0.9855	0.0146
2	**0.9440**	**0.0561**
3	0.8822	0.1179
4	0.8080	0.1920
5	0.7292	0.2708
6	0.6516	0.3484
7	0.5787	0.4213
8	0.5126	0.4874
9	0.4539	0.5461
10	0.4023	0.5976

Table 13: Full Sensitivity Analysis Results using Sobol Method

Date	Team 1	Team 2	Score
19 Sep 2023	Galatasaray	FC Kobenhavn	2-2
19 Sep 2023	Bayern Munchen	Manchester United	4-3
19 Sep 2023	Manchester United	Galatasaray	2-3
19 Sep 2023	FC Kobenhavn	Bayern Munchen	1-2
19 Sep 2023	Galatasaray	Bayern Munchen	1-3
19 Sep 2023	Manchester United	FC Kobenhavn	1-0
19 Sep 2023	Bayern Munchen	Galatasaray	2-1
19 Sep 2023	FC Kobenhavn	Manchester United	4-3
20 Sep 2023	Galatasaray	Manchester United	3-3
20 Sep 2023	Bayern Munchen	FC Kobenhavn	0-0
20 Sep 2023	Manchester United	Bayern Munchen	0-1
20 Sep 2023	FC Kobenhavn	Galatasaray	1-0
20 Sep 2023	Arsenal	PSV Eindhoven	4-0
20 Sep 2023	Sevilla	RC Lens	1-1
20 Sep 2023	PSV Eindhoven	Sevilla	2-2
20 Sep 2023	RC Lens	Arsenal	2-1
03 Oct 2023	RC Lens	PSV Eindhoven	1-1
03 Oct 2023	Sevilla	Arsenal	1-2
03 Oct 2023	Arsenal	Sevilla	2-0
03 Oct 2023	PSV Eindhoven	RC Lens	1-0
03 Oct 2023	Sevilla	PSV Eindhoven	2-3
03 Oct 2023	Arsenal	RC Lens	6-0
03 Oct 2023	PSV Eindhoven	Arsenal	1-1
03 Oct 2023	RC Lens	Sevilla	2-1
04 Oct 2023	Real Madrid	1. FC Union Berlin	1-0
04 Oct 2023	Sporting Braga	Napoli	1-2
04 Oct 2023	1. FC Union Berlin	Sporting Braga	2-3
04 Oct 2023	Napoli	Real Madrid	2-3

04 Oct 2023	Sporting Braga	Real Madrid	1-2
04 Oct 2023	1. FC Union Berlin	Napoli	0-1
04 Oct 2023	Napoli	1. FC Union Berlin	1-1
04 Oct 2023	Real Madrid	Sporting Braga	3-0
24 Oct 2023	Sporting Braga	1. FC Union Berlin	1-1
24 Oct 2023	Real Madrid	Napoli	4-2
24 Oct 2023	1. FC Union Berlin	Real Madrid	2-3
24 Oct 2023	Napoli	Sporting Braga	2-0
24 Oct 2023	Benfica	Red Bull Salzburg	0-2
24 Oct 2023	Real Sociedad	Internazionale	1-1
24 Oct 2023	Red Bull Salzburg	Real Sociedad	2-0
24 Oct 2023	Internazionale	Benfica	1-0
25 Oct 2023	Internazionale	Red Bull Salzburg	2-1
25 Oct 2023	Benfica	Real Sociedad	0-1
25 Oct 2023	Real Sociedad	Benfica	3-1
25 Oct 2023	Red Bull Salzburg	Internazionale	0-1
25 Oct 2023	Real Sociedad	Red Bull Salzburg	0-0
25 Oct 2023	Internazionale	Real Sociedad	0-0
25 Oct 2023	Red Bull Salzburg	Benfica	3-1
25 Oct 2023	Feyenoord	Celtic	2-0
07 Nov 2023	Lazio	Atletico Madrid	1-1
07 Nov 2023	Atletico Madrid	Feyenoord	3-2
07 Nov 2023	Celtic	Lazio	1-2
07 Nov 2023	Feyenoord	Lazio	3-1
07 Nov 2023	Celtic	Atletico Madrid	2-2
07 Nov 2023	Atletico Madrid	Celtic	6-0
07 Nov 2023	Lazio	Feyenoord	1-0
07 Nov 2023	Lazio	Celtic	2-0
08 Nov 2023	Feyenoord	Atletico Madrid	1-3
08 Nov 2023	Celtic	Feyenoord	2-1
08 Nov 2023	Atletico Madrid	Lazio	2-0
08 Nov 2023	AC Milan	Newcastle United	0-0
08 Nov 2023	Paris St Germain	Borussia Dortmund	2-0
08 Nov 2023	Newcastle United	Paris St Germain	4-1
08 Nov 2023	Borussia Dortmund	AC Milan	0-0
08 Nov 2023	Newcastle United	Borussia Dortmund	1-0
28 Nov 2023	Paris St Germain	AC Milan	3-0
28 Nov 2023	Borussia Dortmund	Newcastle United	2-0
28 Nov 2023	AC Milan	Paris St Germain	2-1
28 Nov 2023	AC Milan	Borussia Dortmund	1-3
28 Nov 2023	Paris St Germain	Newcastle United	1-1
28 Nov 2023	Newcastle United	AC Milan	1-2
28 Nov 2023	Borussia Dortmund	Paris St Germain	1-1
28 Nov 2023	Young Boys Berne	RB Leipzig	1-3
29 Nov 2023	Manchester City	Crvena Zvezda	3-1
29 Nov 2023	Crvena Zvezda	Young Boys Berne	2-2
29 Nov 2023	RB Leipzig	Manchester City	1-3
29 Nov 2023	Young Boys Berne	Manchester City	1-3
29 Nov 2023	RB Leipzig	Crvena Zvezda	3-1
29 Nov 2023	Manchester City	Young Boys Berne	3-0

29 Nov 2023	Crvena Zvezda	RB Leipzig	1-2
12 Dec 2023	Manchester City	RB Leipzig	3-2
12 Dec 2023	Young Boys Berne	Crvena Zvezda	2-0
12 Dec 2023	Crvena Zvezda	Manchester City	2-3
12 Dec 2023	RB Leipzig	Young Boys Berne	2-1
12 Dec 2023	Barcelona	Royal Antwerp	5-0
12 Dec 2023	Shakhtar Donetsk	FC Porto	1-3
12 Dec 2023	Royal Antwerp	Shakhtar Donetsk	2-3
12 Dec 2023	FC Porto	Barcelona	0-1
13 Dec 2023	Barcelona	Shakhtar Donetsk	2-1
13 Dec 2023	Royal Antwerp	FC Porto	1-4
13 Dec 2023	Shakhtar Donetsk	Barcelona	1-0
13 Dec 2023	FC Porto	Royal Antwerp	2-0
13 Dec 2023	Shakhtar Donetsk	Royal Antwerp	1-0
13 Dec 2023	Barcelona	FC Porto	2-1
13 Dec 2023	FC Porto	Shakhtar Donetsk	3-5
13 Dec 2023	Royal Antwerp	Barcelona	3-2

Table 14: UEFA Group Stage Match Data, 2023-2024

Date	Team 1	Team 2	Score	Penalty
13 Feb 2024	FC Kobenhavn	Manchester City	1-3	-
13 Feb 2024	RB Leipzig	Real Madrid	0-1	-
14 Feb 2024	Lazio	Bayern Munchen	1-0	-
14 Feb 2024	Paris St Germain	Real Sociedad	2-0	-
20 Feb 2024	Internazionale	Atletico Madrid	1-0	-
20 Feb 2024	PSV Eindhoven	Borussia Dortmund	1-1	-
21 Feb 2024	FC Porto	Arsenal	1-0	-
21 Feb 2024	Napoli	Barcelona	1-1	-
05 Mar 2024	Bayern Munchen	Lazio	3-0	-
05 Mar 2024	Real Sociedad	Paris St Germain	1-2	-
06 Mar 2024	Manchester City	FC Kobenhavn	3-1	-
06 Mar 2024	Real Madrid	RB Leipzig	1-1	-
12 Mar 2024	Arsenal	FC Porto	1-0	4-2
12 Mar 2024	Barcelona	Napoli	3-1	-
13 Mar 2024	Atletico Madrid	Internazionale	2-1	5-4
13 Mar 2024	Borussia Dortmund	PSV Eindhoven	2-0	-
09 Apr 2024	Arsenal	Bayern Munchen	2-2	-
09 Apr 2024	Real Madrid	Manchester City	3-3	-
10 Apr 2024	Atletico Madrid	Borussia Dortmund	2-1	-
10 Apr 2024	Paris St Germain	Barcelona	2-3	-
16 Apr 2024	Barcelona	Paris St Germain	1-4	-
16 Apr 2024	Borussia Dortmund	Atletico Madrid	4-2	-
17 Apr 2024	Bayern Munchen	Arsenal	1-0	-
17 Apr 2024	Manchester City	Real Madrid	1-1	3-4
30 Apr 2024	Bayern Munchen	Real Madrid	2-2	-
01 May 2024	Borussia Dortmund	Paris St Germain	0-1	-
07 May 2024	Paris St Germain	Borussia Dortmund	0-1	-
08 May 2024	Real Madrid	Bayern Munchen	2-1	-

01 Jun 2024	Borussia Dortmund	Real Madrid	0-2	-

Table 15: UEFA Playoffs Match Data, 2023-2024

Date	**Team 1**	**Team 2**	**Score**
02 Apr 2024	Cobresal	Barcelona	1-1
02 Apr 2024	The Strongest	Grêmio	2-0
02 Apr 2024	Universitario	LDU Quito	2-1
02 Apr 2024	Millonarios	Flamengo	1-1
02 Apr 2024	Deportivo Táchira	River Plate	0-2
3 Apr 2024	Colo-Colo	Cerro Porteño	1-0
3 Apr 2024	Alianza Lima	Fluminse	1-1
03 Apr 2024	Huachipato	Estudiantes	1-1
03 Apr 2024	Botafogo	Junior	1-3
03 Apr 2024	San Lorenzo	Palmeiras	1-1
03 Apr 2024	Nacional	Libertad	2-0
04 Apr 2024	Talleres	São Paulo	2-1
04 Apr 2024	Palestino	Bolívar	0-4
04 Apr 2024	Liverpool	Independiente del Valle	1-1
04 Apr 2024	Caracas	Atlético Mineiro	1-4
04 Apr 2024	Rosario Central	Peñarol	1-0
9 Apr 2024	Fluminense	Colo-Colo	2-1
09 Apr 2024	Estudiantes	The Strongest	2-1
09 Apr 2024	Junior	Universitario	1-1
09 Apr 2024	Libertad	Deportivo Táchira	3-0
10 Apr 2024	Cerro Porteño	Alianza Lima	1-0
10 Apr 2024	São Paulo	Cobresal	2-0
10 Apr 2024	Barcelona	Talleres	2-2
10 Apr 2024	Flamengo	Palestino	2-0
10 Apr 2024	Independiente del Valle	San Lorenzo	2-0
10 Apr 2024	Atlético Mineiro	Rosario Central	2-1
10 Apr 2024	Peñarol	Caracas	5-0
11 Apr 2024	LDU Quito	Botafogo	1-0
11 Apr 2024	Bolívar	Millonarios	3-2
11 Apr 2024	River Plate	Nacional	2-0
21 Apr 2024	Grêmio	Huachipato	0-2
21 Apr 2024	Palmeiras	Liverpool	3-1
23 Apr 2024	Colo-Colo	Alianza Lima	0-0
23 Apr 2024	Estudiantes	Grêmio	0-1
23 Apr 2024	Junior	LDU Quito	1-1
23 Apr 2024	Liverpool	San Lorenzo	1-0
23 Apr 2024	Caracas	Rosario Central	1-1
23 Apr 2024	Atlético Mineiro	Peñarol	3-2
24 Apr 2024	Huachipato	The Strongest	0-0
24 Apr 2024	Botafogo	Universitario	3-1
24 Apr 2024	Bolívar	Flamengo	2-1
24 Apr 2024	Independiente del Valle	Palmeiras	2-3
24 Apr 2024	Nacional	Deportivo Táchira	2-1
24 Apr 2024	Libertad	River Plate	1-2

25 Apr 2024	Cerro Porteño	Fluminse	0-0
25 Apr 2024	Cobresal	Talleres	0-2
25 Apr 2024	Barcelona	São Paulo	0-2
25 Apr 2024	Palestino	Millonarios	3-1
07 May 2024	Universitario	Junior	1-1
07 May 2024	Palestino	Flamengo	1-0
07 May 2024	Rosario Central	Atlético Mineiro	0-1
07 May 2024	Caracas	Peñarol	0-1
07 May 2024	Deportivo Táchira	Libertad	1-1
07 May 2024	Nacional	River Plate	2-2
08 May 2024	Alianza Lima	Cerro Porteño	1-1
08 May 2024	Talleres	Barcelona	3-1
08 May 2024	Cobresal	São Paulo	1-3
08 May 2024	Botafogo	LDU Quito	2-1
08 May 2024	Millonarios	Bolívar	1-1
09 May 2024	Colo-Colo	Fluminense	0-1
09 May 2024	The Strongest	Estudiantes	1-0
09 May 2024	San Lorenzo	Independiente del Valle	2-0
09 May 2024	Liverpool	Palmeiras	0-5
14 May 2024	Talleres	Cobresal	1-0
14 May 2024	LDU Quito	Junior	0-1
14 May 2024	Millonarios	Palestino	1-1
14 May 2024	Peñarol	Atlético Mineiro	2-0
14 May 2024	River Plate	Libertad	2-0
15 May 2024	Alianza Lima	Colo-Colo	1-1
15 May 2024	The Strongest	Huachipato	4-0
15 May 2024	Flamengo	Bolívar	4-0
15 May 2024	Palmeiras	Independiente del Valle	2-1
15 May 2024	Deportivo Táchira	Nacional	0-1
16 May 2024	Fluminense	Cerro Porteño	2-1
16 May 2024	São Paulo	Barcelona	0-0
16 May 2024	Universitario	Botafogo	0-1
16 May 2024	San Lorenzo	Liverpool	3-2
16 May 2024	Rosario Central	Caracas	4-1
28 May 2024	LDU Quito	Universitario	2-0
28 May 2024	Junior	Botafogo	0-0
28 May 2024	Flamengo	Millonarios	3-0
28 May 2024	Bolívar	Palestino	3-1
28 May 2024	Peñarol	Rosario Central	2-1
28 May 2024	Atlético Mineiro	Caracas	4-0
29 May 2024	Fluminense	Alianza Lima	3-2
29 May 2024	Cerro Porteño	Colo-Colo	1-1
29 May 2024	São Paulo	Talleres	2-0
29 May 2024	Barcelona	Cobresal	2-1
29 May 2024	Grêmio	The Strongest	4-0
29 May 2024	Estudiantes	Huachipato	3-4
30 May 2024	Palmeiras	San Lorenzo	0-0
30 May 2024	Independiente del Valle	Liverpool	2-1
30 May 2024	River Plate	Deportivo Táchira	2-0
30 May 2024	Libertad	Nacional	2-1

04 Jun 2024	Huachipato	Grêmio	0-1
08 Jun 2024	Grêmio	Estudiantes	1-1

Table 16: CONMEBOL Groups Match Data, 2023-2024

Team 1	Team 2	Score	Penalty
San Lorenzo	Atlético Mineiro	1-2	-
Nacional	São Paulo	0-2	-
Flamengo	Bolívar	2-1	-
Colo-Colo	Junior	3-1	-
Talleres	River Plate	1-3	-
Peñarol	The Strongest	4-1	-
Botafogo	Palmeiras	4-3	-
Grêmio	Fluminense	3-3	2-4
Fluminense	Atlético Mineiro	1-2	-
Botafogo	São Paulo	1-1	5-4
Flamengo	Peñarol	0-2	-
Colo-Colo	River Plate	1-2	-
Atlético Mineiro	River Plate	3-0	-
Botafogo	Peñarol	6-3	-
Atlético Mineiro	Botafogo	1-3	-

Table 17: CONMEBOL Playoffs Match Data, 2023-2024

No.	Team 1	Team 2	Score	Away
1	Saprissa	Philadelphia Union	5-6	0
2	Herediano	Toluca	4-4	1
3	Real Estelí	América	2-3	0
4	Forge FC	Guadalajara	0-5	0
5	Independiente	New England Revolution	0-4	0
6	St. Louis City SC	Houston Dynamo FC	2-2	1
7	Vancouver Whitecaps FC	UANL	1-4	0
8	Cavalry FC	Orlando City SC	1-6	0
9	Comunicaciones	Monterrey	1-7	0
10	Cavalier	FC Cincinnati	0-6	0
11	Moca	Nashville SC	0-7	0

Table 18: CONCACAF Group Stage Match Data, 2023-2024

No	Team 1	Team 2	Score	Penalty
1	Philadelphia Union	Pachuca	0-6	-
2	Herediano	Robinhood	3-1	-
3	Guadalajara	América	3-5	-
4	New England Revolution	Alajuelense	5-1	-
5	Houston Dynamo FC	Columbus Crew	1-2	-
6	Orlando City SC	UANL	2-4	-
7	FC Cincinnati	Monterrey	1-3	-
8	Nashville SC	Inter Miami CF	3-5	-

1	Herediano	Pachuca	1-7	-
2	New England Revolution	América	2-9	-
3	Columbus Crew	UANL	2-2	4-3
4	Inter Miami CF	Monterrey	2-5	-
1	América	Pachuca	2-3	-
2	Columbus Crew	Monterrey	5-2	-
1	Pachuca	Columbus Crew	3-0	-

Table 19: CONCACAF Playoffs Match Data

No	Date	Team 1	Team 2	Score
1	18 Sep 2023	Al-Faisaly	Nasaf	0-1
2	18 Sep 2023	Al Sadd	Sharjah	0-0
1	18 Sep 2023	Al-Ittihad	AGMK	3-0
2	18 Sep 2023	Al-Quwa Al-Jawiya	Sepahan	2-2
1	18 Sep 2023	Al-Hilal	Navbahor	1-1
2	18 Sep 2023	Mumbai City	Nassaji Mazandaran	0-2
1	19 Sep 2023	Pakhtakor	Al-Ain	0-3
2	19 Sep 2023	Ahal	Al-Fayha	1-0
1	19 Sep 2023	Istiklol	Al-Duhail	0-0
2	19 Sep 2023	Persepolis	Al Nassr	0-2
1	19 Sep 2023	Yokohama F. Marinos	Incheon United	2-4
2	19 Sep 2023	Kaya-Iloilo	Shandong Taishan	1-3
1	19 Sep 2023	Ulsan Hyundai	BG Pathum United	3-1
2	19 Sep 2023	Johor Darul Ta'zim	Kawasaki Frontale	0-1
1	20 Sep 2023	Jeonbuk Hyundai Motors	Kitchee	2-1
2	20 Sep 2023	Lion City Sailors	Bangkok United	1-2
1	20 Sep 2023	Buriram United	Zhejiang	4-1
2	20 Sep 2023	Melbourne City	Ventforet Kofu	0-0
1	20 Sep 2023	Wuhan Three Towns	Urawa Red Diamonds	2-2
2	20 Sep 2023	Hanoi FC	Pohang Steelers	2-4
3	02 Oct 2023	Sharjah	Al-Faisaly	1-0
4	02 Oct 2023	Nasaf	Al Sadd	3-1
3	02 Oct 2023	AGMK	Al-Quwa Al-Jawiya	1-2
4	02 Oct 2023	Nasaf	Al-Ittihad	3-1
3	02 Oct 2023	Navbahor	Mumbai City	3-0
3	02 Oct 2023	Al Nassr	Istiklol	3-1
4	02 Oct 2023	Al-Duhail	Persepolis	0-1
3	03 Oct 2023	Al-Ain	Ahal	4-2
4	03 Oct 2023	Al-Fayha	Pakhtakor	2-0
4	03 Oct 2023	Nassaji Mazandaran	Al-Hilal	0-3
3	03 Oct 2023	Incheon United	Kaya-Iloilo	4-0
4	03 Oct 2023	Shandong Taishan	Yokohama F. Marinos	0-1
3	03 Oct 2023	Kawasaki Frontale	Ulsan Hyundai	1-0
4	03 Oct 2023	BG Pathum United	Johor Darul Ta'zim	2-4
3	04 Oct 2023	Bangkok United	Jeonbuk Hyundai Motors	3-2
4	04 Oct 2023	Kitchee	Lion City Sailors	1-2
3	04 Oct 2023	Zhejiang	Melbourne City	1-2
4	04 Oct 2023	Ventforet Kofu	Buriram United	1-0

3	04 Oct 2023	Urawa Red Diamonds	Hanoi FC	6-0
4	04 Oct 2023	Pohang Steelers	Wuhan Three Towns	3-1
5	23 Oct 2023	Sharjah	Nasaf	1-0
6	23 Oct 2023	Al Sadd	Al-Faisaly	6-0
5	23 Oct 2023	Al-Quwa Al-Jawiya	Al-Ittihad	2-0
6	23 Oct 2023	AGMK	Sepahan	1-3
5	23 Oct 2023	Navbahor	Nassaji Mazandaran	2-1
6	23 Oct 2023	Al-Hilal	Mumbai City	6-0
5	24 Oct 2023	Al-Ain	Al-Fayha	4-1
6	24 Oct 2023	Pakhtakor	Ahal	3-0
5	24 Oct 2023	Al Nassr	Al-Duhail	4-3
6	24 Oct 2023	Persepolis	Istiklol	2-0
5	24 Oct 2023	Ulsan Hyundai	Johor Darul Ta'zim	3-1
6	24 Oct 2023	BG Pathum United	Kawasaki Frontale	2-4
5	25 Oct 2023	Jeonbuk Hyundai Motors	Lion City Sailors	3-0
6	25 Oct 2023	Kitchee	Bangkok United	2-1
5	25 Oct 2023	Incheon United	Shandong Taishan	0-2
6	25 Oct 2023	Yokohama F. Marinos	Kaya-Iloilo	3-0
5	25 Oct 2023	Zhejiang	Ventforet Kofu	2-0
6	25 Oct 2023	Buriram United	Melbourne City	0-2
5	25 Oct 2023	Urawa Red Diamonds	Pohang Steelers	0-2
6	25 Oct 2023	Wuhan Three Towns	Hanoi FC	2-1
7	06 Nov 2023	Nasaf	Sharjah	1-1
8	06 Nov 2023	Al-Faisaly	Al Sadd	2-0
7	06 Nov 2023	Sepahan	AGMK	9-0
8	06 Nov 2023	Al-Quwa Al-Jawiya	Al-Ittihad	2-0
7	06 Nov 2023	Nassaji Mazandaran	Navbahor	1-3
8	06 Nov 2023	Mumbai City	Al-Hilal	0-2
7	07 Nov 2023	Al-Fayha	Al-Ain	2-3
8	07 Nov 2023	Ahal	Pakhtakor	1-1
7	07 Nov 2023	Al-Duhail	Al Nassr	2-3
8	07 Nov 2023	Istiklol	Persepolis	1-1
1	07 Nov 2023	Shandong Taishan	Incheon United	3-1
2	07 Nov 2023	Kaya-Iloilo	Yokohama F. Marinos	1-2
1	07 Nov 2023	Kawasaki Frontale	BG Pathum United	4-2
2	07 Nov 2023	Johor Darul Ta'zim	Ulsan Hyundai	2-1
7	08 Nov 2023	Lion City Sailors	Jeonbuk Hyundai Motors	2-0
8	08 Nov 2023	Bangkok United	Kitchee	1-1
7	08 Nov 2023	Ventforet Kofu	Zhejiang	4-1
8	08 Nov 2023	Melbourne City	Buriram United	0-1
7	08 Nov 2023	Pohang Steelers	Urawa Red Diamonds	2-1
8	08 Nov 2023	Hanoi FC	Wuhan Three Towns	2-1
9	27 Nov 2023	Nasaf	Al-Faisaly	3-1
10	27 Nov 2023	Sharjah	Al Sadd	0-2
9	27 Nov 2023	AGMK	Al-Ittihad	1-2
10	27 Nov 2023	Sepahan	Al-Quwa Al-Jawiya	1-0
9	27 Nov 2023	Al Nassr	Persepolis	0-0
10	27 Nov 2023	Al-Duhail	Istiklol	2-0
9	28 Nov 2023	Al-Ain	Pakhtakor	1-3
10	28 Nov 2023	Al-Fayha	Ahal	3-0

9	28 Nov 2023	Navbahor	Al-Hilal	0-2
10	28 Nov 2023	Nassaji Mazandaran	Mumbai City	2-0
3	28 Nov 2023	Incheon United	Yokohama F. Marinos	2-1
4	28 Nov 2023	Shandong Taishan	Kaya-Iloilo	6-1
3	28 Nov 2023	BG Pathum United	Ulsan Hyundai	1-3
4	28 Nov 2023	Kawasaki Frontale	Johor Darul Ta'zim	5-0
9	29 Nov 2023	Kitchee	Jeonbuk Hyundai Motors	1-2
10	29 Nov 2023	Bangkok United	Lion City Sailors	1-0
9	29 Nov 2023	Zhejiang	Buriram United	3-2
10	29 Nov 2023	Ventforet Kofu	Melbourne City	3-3
9	29 Nov 2023	Urawa Red Diamonds	Wuhan Three Towns	2-1
10	29 Nov 2023	Pohang Steelers	Hanoi FC	2-0
11	04 Dec 2023	Al-Faisaly	Sharjah	2-1
12	04 Dec 2023	Al Sadd	Nasaf	2-2
11	04 Dec 2023	Al-Ittihad	Sepahan	2-1
12	04 Dec 2023	Al-Quwa Al-Jawiya	AGMK	3-2
11	04 Dec 2023	Al-Hilal	Nassaji Mazandaran	2-1
12	04 Dec 2023	Mumbai City	Navbahor	1-2
11	05 Dec 2023	Pakhtakor	Al-Fayha	1-4
12	05 Dec 2023	Ahal	Al-Ain	1-4
11	05 Dec 2023	Persepolis	Al-Duhail	1-2
12	05 Dec 2023	Istiklol	Al Nassr	1-1
11	06 Dec 2023	Wuhan Three Towns	Pohang Steelers	1-2
12	06 Dec 2023	Hanoi FC	Urawa Red Diamonds	2-1
11	12 Dec 2023	Buriram United	Ventforet Kofu	2-3
12	12 Dec 2023	Melbourne City	Zhejiang	1-1
5	12 Dec 2023	Ulsan Hyundai	Kawasaki Frontale	2-2
6	12 Dec 2023	Johor Darul Ta'zim	BG Pathum United	4-1
11	13 Dec 2023	Lion City Sailors	Kitchee	0-2
12	13 Dec 2023	Jeonbuk Hyundai Motors	Bangkok United	3-2
5	13 Dec 2023	Yokohama F. Marinos	Shandong Taishan	3-0
6	13 Dec 2023	Kaya-Iloilo	Incheon United	1-3

Table 20: AFC Group Stage Match Data, 2023-2024

No	**Team 1**	Team 2	Score	Penalty
1	Nasaf	Al Ain	1-2	-
2	Al-Fayha	Al Nassr	0-3	-
3	Sepahan	Al Hilal	2-6	-
4	Navbahor	Al-Ittihad	1-2	-
5	Jeonbuk Hyundai Motors	Pohang Steelers	3-1	-
6	Ulsan Hyundai	Ventforet Kofu	5-1	-
7	Shandong Taishan	Kawasaki Frontale	6-5	-
8	Bangkok United	Yokohama F. Marinos	2-3	-
9	Al Ain	Al Nassr	4-4	3-1
10	Al Hilal	Al-Ittihad	4-0	-
11	Jeonbuk Hyundai Motors	Ulsan Hyundai	1-2	-
12	Shandong Taishan	Yokohama F. Marinos	1-3	-
13	Al Ain	Al Hilal	5-4	-

14	Ulsan Hyundai	Yokohama F. Marinos	3-3	4-5
15	Yokohama F. Marinos	Al Ain	3-6	-

Table 21: AFC Playoffs Match Data, 2023-2024

No.	**Date**	**Team 1**	**Team 2**	**Score**
1	11 May 2024	Rewa	Auckland City	2-2
2	11 May 2024	Solomon Warriors	Hekari United	0-2
7	12 May 2024	Pirae	Vaivase-Tai FC	6-0
3	14 May 2024	Auckland City	Hekari United	1-0
4	14 May 2024	Solomon Warriors	Rewa	2-3
9	15 May 2024	AS Magenta	Vaivase-Tai FC	8-0
10	16 May 2024	Pirae	Ifira Black Bird	5-1
5	17 May 2024	Hekari United	Rewa	2-3
6	17 May 2024	Auckland City	Solomon Warriors	5-0
11	18 May 2024	Vaivase-Tai FC	Ifira Black Bird	1-5
12	18 May 2024	AS Magenta	Pirae	0-2
8	20 May 2024	Ifira Black Bird	AS Magenta	1-2

Table 22: OFC Group Stage Match Data, 2023-2024

Date	**Team 1**	**Team 2**	Score	Penalty
22 May 2024	Auckland City	AS Magenta	1-0	-
22 May 2024	Pirae	Rewa	4-2	-
24 May 2024	Auckland City	Pirae	4-0	-

Table 23: OFC Playoffs Match Data, 2023-2024

Date	**Team 1**	**Team 2**	**Score**
25 Nov 2023	CR Belouizdad	Young Africans	3-0
25 Nov 2023	Simba SC	ASEC Mimosas	1-1
25 Nov 2023	Pyramids FC	TP Mazembe	1-0
26 Nov 2023	Al Ahly	Medeama	3-0
26 Nov 2023	Petro de Luanda	Al-Hilal Omdurman	1-0
26 Nov 2023	ES Tunis	Étoile du Sahel	2-0
26 Nov 2023	Wydad Casablanca	Jwaneng Galaxy	0-1
26 Nov 2023	Mamelodi Sundowns	Nouadhibou	3-0
02 Dec 2023	CR Belouizdad	Medeama	2-1
02 Dec 2023	Al-Hilal Omdurman	ES Tunis	3-1
02 Dec 2023	Jwaneng Galaxy	Simba SC	0-0
02 Dec 2023	TP Mazembe	Mamelodi Sundowns	1-0
03 Dec 2023	Young Africans	Al Ahly	1-1
03 Dec 2023	Étoile du Sahel	Petro de Luanda	0-2
03 Dec 2023	ASEC Mimosas	Wydad Casablanca	1-0
03 Dec 2023	Nouadhibou	TP Mazembe	2-0
09 Dec 2023	Medeama	Young Africans	1-1
09 Dec 2023	Al Ahly	CR Belouizdad	0-0
09 Dec 2023	Étoile du Sahel	Al-Hilal Omdurman	1-0

09 Dec 2023	Jwaneng Galaxy	Wydad Casablanca	0-2
10 Dec 2023	ES Tunis	Petro de Luanda	0-0
10 Dec 2023	Wydad Casablanca	Simba SC	1-0
10 Dec 2023	Mamelodi Sundowns	Pyramids FC	0-0
10 Dec 2023	Nouadhibou	TP Mazembe	0-0
19 Dec 2023	Simba SC	Wydad Casablanca	2-0
19 Dec 2023	TP Mazembe	Nouadhibou	2-0
20 Dec 2023	Young Africans	Medeama	3-0
20 Dec 2023	Al-Hilal Omdurman	Étoile du Sahel	1-1
20 Dec 2023	Petro de Luanda	ES Tunis	0-0
20 Dec 2023	ASEC Mimosas	Jwaneng Galaxy	3-0
20 Dec 2023	Pyramids FC	Mamelodi Sundowns	0-1
17 Feb 2024	CR Belouizdad	Al Ahly	0-0
24 Feb 2024	Medeama	Al Ahly	0-1
24 Feb 2024	Al-Hilal Omdurman	Petro de Luanda	0-0
24 Feb 2024	Jwaneng Galaxy	Wydad Casablanca	0-1
24 Feb 2024	ASEC Mimosas	Simba SC	0-0
24 Feb 2024	TP Mazembe	Pyramids FC	3-0
25 Feb 2024	Young Africans	CR Belouizdad	4-0
25 Feb 2024	Étoile du Sahel	ES Tunis	0-2
25 Feb 2024	Nouadhibou	Mamelodi Sundowns	0-2
02 Mar 2024	CR Belouizdad	Medeama	3-0
02 Mar 2024	Al Ahly	Young Africans	1-0
02 Mar 2024	Pyramids FC	Nouadhibou	2-2
02 Mar 2024	Mamelodi Sundowns	TP Mazembe	1-0
03 Mar 2024	Petro de Luanda	Étoile du Sahel	2-0
03 Mar 2024	ES Tunis	Al-Hilal Omdurman	1-0
03 Mar 2024	Wydad Casablanca	ASEC Mimosas	1-0
03 Mar 2024	Simba SC	Jwaneng Galaxy	6-0

Table 24: CAF Group Stage Match Data

Team 1	**Team 2**	Score	Penalty
Simba SC	Al Ahly	0-3	-
Petro de Luanda	TP Mazembe	1-2	-
ES Tunis	ASEC Mimosas	0-0	4-2
Young Africans	Mamelodi Sundowns	0-0	3-2
ES Tunis	Mamelodi Sundowns	2-0	-
TP Mazembe	Al Ahly	0-3	-
ES Tunis	Al Ahly	0-1	-

Table 25: CAF Playoffs Match Data, 2023-2024

Group A

Cost: 88028

```
[[-4 -5  4  2 -3 -6 -2  3  5  6]
 [ 5 -6  3 -1  6 -4  1  4 -3 -5]
 [ 6 -4 -2  5  1 -5 -6 -1  2  4]
 [ 1  3 -1 -6 -5  2  5 -2  6 -3]
 [-2  1  6 -3  4  3 -4 -6 -1  2]
 [-3  2 -5  4 -2  1  3  5 -4 -1]]
```

Group B

Cost: 127329

```
[[ 4 -2  3  2 -3  6 -4 -5 -6  5]
 [-3  1 -6 -1  6  4 -5  3  5 -4]
 [ 2  5 -1 -4  1 -5 -6 -2  4  6]
 [-1 -6 -5  3  5 -2  1  6 -3  2]
 [-6 -3  4  6 -4  3  2  1 -2 -1]
 [ 5  4  2 -5 -2 -1  3 -4  1 -3]]
```

Group C

Cost = 233922.0

```
[[ 2 -4 -5  6  4 -3  5 -6 -2  3]
 [-1 -3 -4  3  6  5 -6  4  1 -5]
 [ 6  2 -6 -2 -5  1 -4  5  4 -1]
 [ 5  1  2 -5 -1 -6  3 -2 -3  6]
 [-4 -6  1  4  3 -2 -1 -3  6  2]
 [-3  5  3 -1 -2  4  2  1 -5 -4]]
```

Group D

Cost = 58929.0

```
[[ 6  4  5 -6 -5  3  2 -4 -2 -3]
 [-5 -6  3  4  6 -4 -1 -3  1  5]
 [-4 -5 -2  5  4 -1 -6  2  6  1]
 [ 3 -1  6 -2 -3  2  5  1 -5 -6]
 [ 2  3 -1 -3  1  6 -4 -6  4 -2]
 [-1  2 -4  1 -2 -5  3  5 -3  4]]
```

Figure 2: Matrices formed from Simulated Annealing

8.2 Appendix B (Code)

8.2.1 Generating Incompability Matrix

```python
import pandas as pd
import numpy as np
from haversine import haversine, Unit



def calculate_travel_cost(team1, team2):
    """
    Calculate travel cost based on ELO difference and geographic distance
    """
    # Calculate ELO difference
    elo_diff = abs(team1['Real ELO'] - team2['Real ELO'])
    print(team1.keys)

    # haversine formula
    point_1 = team1['Latitude'], team1['Longitude']
    point_2 = team2['Latitude'], team2['Longitude']
    geo_distance = haversine(point_1, point_2)

    return elo_diff*elo_diff + 4*geo_distance

```

```python

def create_incompability_matrix(df):
    """
    Create a incompability matrix for double round-robin tournament
    """
    # Number of teams
    n = len(df)

    # Initialize incompability matrix with string indicators
    incompability_matrix = np.full((n, n), '-', dtype=object)
    np.fill_diagonal(incompability_matrix, 0)

    # Calculate costs for all pair combinations
    for i in range(n):
        for j in range(n):
            if i != j:
                incompability_matrix[i, j] = round(calculate_travel_cost(df.iloc[i], df.iloc[j]), 2)

    matrix_df = pd.DataFrame(
        incompability_matrix,
        columns=df['Teams'],
        index=df['Teams']
    )

    return matrix_df


# Read the CSV data
df = pd.read_csv('file.csv')
df = df.drop_duplicates(subset=['Teams'])

# Calculate and display the incompability matrix
incompability_matrix = create_incompability_matrix(df)

# Outputs
print(incompability_matrix)
incompability_matrix.to_csv('incompability.csv')
```

8.2.2 Create Groupings

```python
import csv
import pandas as pd
import ast
import matplotlib.pyplot as plt
# from Group_generator.calculate_cost import distance_matrix


class TeamGroupOptimizer:
    def __init__(self, csv_path, combi_path):

        # Read the CSV file
        df = pd.read_csv(csv_path, index_col=0)

        # Convert to numpy array
        self.distance_matrix = df.to_numpy()
        self.team_names = df.index.tolist()
        self.leader_teams = df.columns.tolist()
        self.group_combi = []

        #Extract all possible combination from file for brute forcing
        with open(combi_path) as f:
            reader = csv.reader(f)
```

```
            for data in reader:
                self.group_combi.append(list(map(lambda x: ast.literal_eval(x), data)))
        print(self.group_combi[:5])


    def calculate_group_metrics(self, group_members):
        """
        Calculate metrics for groups using predefined distances
        """

        # Calculate group distances
        final_group_cost = 0
        for leader_idx, group in enumerate(group_members):

            group_cost = 0

            # Calculate group distance using predefined matrix
            for i, team in enumerate(group):
                group_cost += self.distance_matrix[team, leader_idx]

            final_group_cost += group_cost

        return {
            'named_groups': group_members,
            'group_cost':final_group_cost,
        }

    def optimize_groups(self):
        """
        Optimize team grouping with predetermined group leaders

        :return: Optimal groups and their metrics
        """

        # Find best group configuration
        best_result = None
        min_cost = float('inf')
        costs = []

        for combi in self.group_combi:
            # Calculate metrics
            result = self.calculate_group_metrics(combi)
            costs.append(result['group_cost'])

            # Update best result if variance is lower
            if result['group_cost'] < min_cost:
                min_cost = result['group_cost']
                best_result = result

        groups = [[self.leader_teams[leader]] + [self.team_names[index] for index in group] for leader, group in enumerate(best_result['named_groups'])]
        best_result['named_groups'] = groups
        best_result['costs'] = costs

        return best_result


def main():
    # Path to the CSV file
    csv_path = 'P4_T16.csv'
    combi_path = 'combination.csv'

```

```
    # Create optimizer
    optimizer = TeamGroupOptimizer(csv_path, combi_path)

    # Run optimization
    results = optimizer.optimize_groups()
    # results = 0

    #Print results
    print("Optimal Team Groups:")
    for i, group in enumerate(results['named_groups'], 1):
        print(f"Group {i}: {group}")
    print("\nOptimization Metrics:")
    print(f"Group Cost: {results['group_cost']}")

    #Generate plot
    plt.figure(figsize=(8, 4))  # Set figure size
    x_values = range(len(results['costs']))
    plt.scatter(x_values, results['costs'], marker='o', color='blue', s=0.1)  # Line plot with markers
    plt.title("Incompatibility Scatter Plot")
    plt.xlabel("Index")
    plt.ylabel("Value")
    plt.grid(True)
    plt.show()


if __name__ == "__main__":
    main()
```

8.2.3 TTP with Simulated Annealing

```python
"""
Part 1) Import of libraries and input file
Part 2) Definition of function to generate canonical pattern and initial solution
Part 3) Simulated Annealing
"""

"""
Part 1
"""

# Imports libs from python
import random
import time
import math
import numpy as np
import numexpr

#Input file name here
file = "groupD.txt"

# Variables for Simulated Annealing, set manually
maxP = 100
maxC = 1000
maxR = 10
T = float(400)
beta = 0.98
weight = 4000
teta = 1.04

# Open & extract input file
with open(file) as f:
    distance_matrix = []
    for line in f:
        line = line.split()  # Breaks elements of line whenever there's a space
        if line:  # If line is not blank
            line = [int(i) for i in line]  # Turns each element of line into an integer and
            distance_matrix.append(line)  # adds to bidimensional array

# Number of teams will be length of distance_matrix
numberOfTeams = len(distance_matrix)

# Defines number of rounds
numberOfRounds = 2 * (numberOfTeams - 1)

# Converts array into a numpy array for faster computation. And changes type of variable to float.
distance_matrix = np.array(distance_matrix, dtype=float)

"""
Part 2
"""

# Function to create canonical pattern or 1-factorization
def findCanonicalPattern(numberOfTeams, numberOfRounds):
    x = int(numberOfRounds / 2)
    y = int(numberOfTeams / 2)
    z = 2

    # Creates 3-dimensional array. For example, for 4 teams: x = 2, y = 3, z = 2. So E = [[[0,0],[0,0]],[[0,0],[0,0]],[[0,0],[0,0]]]
```

```
    E = np.zeros((x, y, z))

    for i in range(int(numberOfRounds / 2)):

        E[i][0][:] = [numberOfTeams,
                      i + 1]  # The first edge of a round is the last team (e.g. team 4) playing team i + 1

        for k in range(int(numberOfTeams / 2 - 1)):  # Then to fill the last edges, use functions F1 and F2

            E[i][k + 1][:] = [F1(i + 1, k + 1, numberOfTeams), F2(i + 1, k + 1, numberOfTeams)]

    return (E)


# Defines F1 & F2 find the canonical pattern
def F1(i, k, numberOfTeams):
    if i + k < numberOfTeams:
        return (i + k)
    else:
        return (i + k - numberOfTeams + 1)

def F2(i, k, numberOfTeams):
    if i - k > 0:
        return (i - k)
    else:
        return (i - k + numberOfTeams - 1)


def getInitialSolution(numberOfTeams):

    solution = np.zeros((numberOfTeams, numberOfRounds), dtype=int)

    # Generate possible round-robin match
    games = findCanonicalPattern(numberOfTeams, numberOfRounds)

    # Creates first half of the tournament
    for i in range(int(numberOfRounds / 2)):

        for k in range(int(numberOfTeams / 2)):
            # Every edge of the canonical pattern is a game between the two nodes
            edge = games[i][k]

            teamA = int(edge[0])
            teamB = int(edge[1])

            # One team plays at home, one team plays away
            solution[teamA - 1][i] = teamB
            solution[teamB - 1][i] = - teamA

    # To create second half, mirror the first half inverting the signs
    temp = solution.copy()
    temp = -1 * np.roll(temp, numberOfRounds / 2, axis=1)
    solution = solution + temp

    return (solution)


"""
Part 3
"""
```

```python


# Defines function for Simulated Annealing
def simulated_annealing(maxP, maxC, maxR, T, beta, weight, teta):
    # Generates a initial solution
    S = getInitialSolution(numberOfTeams)
    S = np.array(S)

    numberOfViolations = 0

    for i in range(numberOfTeams):

        count = 0

        for k in range(1, numberOfRounds):

            # If you play two consecutive games at home or aways, increment count by 1. Else, make it 0.
            if (S[i][k] > 0 and S[i][k - 1] > 0) or (S[i][k] < 0 and S[i][k - 1] < 0):
                count += 1
            else:
                count = 0

            # three consecutive game.
            if count > 2:
                numberOfViolations += 1

            # same team in a row
            if abs(S[i][k]) == abs(S[i][k - 1]):
                numberOfViolations += 1

    violationsS = numberOfViolations

    # To calculate total distance, sets total distance to 0
    totaldistance = 0

    # Makes a copy of the solution
    distanceS = np.copy(S)
    for x in range(numberOfTeams):

        distanceS[x] = [x + 1 if i > 0 else abs(i) for i in distanceS[x]]


        totaldistance += distance_matrix[x][distanceS[x][0] - 1]

        for y in range(numberOfRounds - 1):
            # add distance between entries
            totaldistance += distance_matrix[distanceS[x][y] - 1][distanceS[x][y + 1] - 1]

        # add distance to total
        totaldistance += distance_matrix[distanceS[x][-1] - 1][x]

    costS = totaldistance

    # If solution is infeasible, penalize it by increasing its cost
    if violationsS != 0:
        costS = math.sqrt(
            (costS ** 2) + (weight * (1 + math.sqrt(violationsS) * math.log(violationsS / float(2)))) ** 2)

    # Sets best feasible solution so far to a very high number
```

```
    bestFeasible = 9999999
    nbf = 9999999

    # Sets best infeasible solution so far to a very high number
    bestInfeasible = 9999999
    nbi = 9999999

    reheat = 0
    counter = 0

    # While system has not reheated maxR times without improving the solution
    while reheat <= maxR:

        phase = 0

        # While system has not decreased the temperature maxP times without improving the solution
        while phase <= maxP:

            counter = 0 #just in case

            # While system has not rejected maxC moves
            while counter <= maxC:

                # Choose random move
                chooseMove = random.randint(0, 4)

                if chooseMove == 0:  # Swap Homes

                    newS = np.copy(S)  # The new solution is a copy of the current one

                    teamA = random.randint(0, numberOfTeams - 1)  # Choose random team A
                    teamB = random.randint(0, numberOfTeams - 1)  # Choose random team B

                    for i in range(numberOfRounds):

                        # When team A and team B play each other
                        if abs(S[teamA][i]) == teamB + 1:
                            # Invert the signs (swap homes)
                            newS[teamA][i] = - S[teamA][i]
                            newS[teamB][i] = - S[teamB][i]


                elif chooseMove == 1:  # Swap Rounds

                    newS = np.copy(S)

                    roundAindex = random.randint(0, numberOfRounds - 1)  # Choose random round A
                    roundBindex = random.randint(0, numberOfRounds - 1)  # Choose random round B

                    # Swap columns of array (swap rounds)
                    newS[:, [roundAindex, roundBindex]] = newS[:, [roundBindex, roundAindex]]


                elif chooseMove == 2:  # Swap Teams

                    newS = np.copy(S)
```

```
                    teamA = random.randint(0, numberOfTeams - 1)  # Choose random team A
                    teamB = random.randint(0, numberOfTeams - 1)  # Choose random team B

                    for i in range(numberOfRounds):

                        if abs(S[teamA][i]) != teamB + 1:  # If team A and team B are not playing each other in round i

                            newS[[teamA, teamB], i] = newS[
                                [teamB, teamA], i]  # Swap their values (swap their opponents)

                            formerAdversaryTeamA = abs(S[teamA][i]) - 1  # Gets former opponent of team A
                            formerAdversaryTeamB = abs(S[teamB][i]) - 1  # Gets former opponent of team B

                            # Now the former opponent of team A plays B at home or away
                            if S[formerAdversaryTeamA][i] > 0:
                                newS[formerAdversaryTeamA][i] = teamB + 1
                            else:
                                newS[formerAdversaryTeamA][i] = -(teamB + 1)

                            # Now the former opponent of team B plays A at home or away
                            if S[formerAdversaryTeamB][i] > 0:
                                newS[formerAdversaryTeamB][i] = teamA + 1
                            else:
                                newS[formerAdversaryTeamB][i] = -(teamA + 1)


                elif chooseMove == 3:  # Partial Swap Rounds

                    newS = np.copy(S)

                    team = random.randint(0, numberOfTeams - 1)  # Choose random team
                    roundAindex = random.randint(0, numberOfRounds - 1)  # Choose random round A
                    roundBindex = random.randint(0, numberOfRounds - 1)  # Choose random round B

                    startCircuit = abs(S[team][roundAindex])
                    finishCircuit = abs(S[team][roundBindex])

                    currentTeam = startCircuit
                    currentRound = roundBindex

                    # Swaps values of Round A and B for the chosen team
                    newS[team, [roundAindex, roundBindex]] = newS[team, [roundBindex, roundAindex]]

                    # Now you must figure out the other teams that you have to swap to fix the schedule, and swap their values
                    while currentTeam != finishCircuit:

                        index = currentTeam - 1

                        newS[index, [roundAindex, roundBindex]] = newS[index, [
```

```
roundBindex, roundAindex]]

                        currentTeam = abs(S[currentTeam - 1][currentRound])

                        if currentRound == roundBindex:
                            currentRound = roundAindex
                        else:
                            currentRound = roundBindex

                    index = currentTeam - 1

                    newS[index, [roundAindex, roundBindex]] = newS[index, [
roundBindex, roundAindex]]


                elif chooseMove == 4:  # Partial Swap Teams

                    newS = np.copy(S)

                    round = random.randint(0, numberOfRounds - 1)
                    teamA = random.randint(0, numberOfTeams - 1)
                    teamB = random.randint(0, numberOfTeams - 1)

                    adversaryA = S[teamA][round]
                    adversaryB = S[teamB][round]

                    # If team A and B are not playing each other, execute the
swap
                    if abs(adversaryB) != teamA + 1:

                        # Swap the teams in the selected round
                        newS[[teamA, teamB], round] = newS[[teamB, teamA], round]

                        affectedTeamA = abs(adversaryA)
                        affectedTeamB = abs(adversaryB)

                        oppositeA = S[affectedTeamA - 1][round]
                        oppositeB = S[affectedTeamB - 1][round]

                        if oppositeA > 0:
                            newS[affectedTeamA - 1][round] = abs(oppositeB)
                        else:
                            newS[affectedTeamA - 1][round] = - abs(oppositeB)

                        if oppositeB > 0:
                            newS[affectedTeamB - 1][round] = abs(oppositeA)
                        else:
                            newS[affectedTeamB - 1][round] = - abs(oppositeA)

                        currentAdversaryB = adversaryB

                        # mirroring problem
                        while currentAdversaryB != adversaryA:

                            currentAdversaryA = currentAdversaryB

                            i = np.nonzero(S[teamA] == currentAdversaryA)[0][0]

                            currentAdversaryB = S[teamB][i]

                            newS[[teamA, teamB], i] = newS[[teamB, teamA], i]

                            affectedTeamA = abs(currentAdversaryA)
```

```
                            affectedTeamB = abs(currentAdversaryB)

                            oppositeA = S[affectedTeamA - 1][i]
                            oppositeB = S[affectedTeamB - 1][i]

                            if oppositeA > 0:
                                newS[affectedTeamA - 1][i] = abs(oppositeB)
                            else:
                                newS[affectedTeamA - 1][i] = - abs(oppositeB)

                            if oppositeB > 0:
                                newS[affectedTeamB - 1][i] = abs(oppositeA)
                            else:
                                newS[affectedTeamB - 1][i] = - abs(oppositeA)

                # initialise violations
                numberOfViolations = 0

                for i in range(numberOfTeams):

                    count = 0

                    for k in range(1, numberOfRounds):

                        if (newS[i][k] > 0 and newS[i][k - 1] > 0) or (newS[i][k] < 0 and newS[i][k - 1] < 0):

                            count += 1

                        else:

                            count = 0

                        if count > 2:
                            numberOfViolations += 1

                        if abs(newS[i][k]) == abs(newS[i][k - 1]):
                            numberOfViolations += 1

                violationsNewS = numberOfViolations

                # get the total distance
                totaldistance = 0
                distanceNewS = np.copy(newS)

                for x in range(numberOfTeams):

                    distanceNewS[x] = [x + 1 if i > 0 else abs(i) for i in distanceNewS[x]]

                    totaldistance += distance_matrix[x][distanceNewS[x][0] - 1]

                    for y in range(numberOfRounds - 1):
                        totaldistance += distance_matrix[distanceNewS[x][y] - 1][distanceNewS[x][y + 1] - 1]

                    totaldistance += distance_matrix[distanceNewS[x][-1] - 1][x]

                costNewS = totaldistance

                # If solution is infeasible, penalize it.
                if violationsNewS != 0:
                    costNewS = math.sqrt((costNewS ** 2) + (
```

```
                                weight * (1 + math.sqrt(violationsNewS) * math.log(violationsNewS / float(2)))) ** 2)

                # If the new solution improve or the best feasible solution or the best infeasible solution
                if costNewS < costS or (violationsNewS == 0 and costNewS < bestFeasible) or (
                        violationsNewS > 0 and costNewS < bestInfeasible):

                    accept = 1

                    if violationsNewS == 0:
                        bestFeasibleSolution = newS.copy()
                    else:
                        bestInfeasibleSolution = newS.copy()

                # Else, accept with a probability given by exp(-delta/T)
                else:

                    delta = float(costNewS - costS)
                    probability = math.exp(-(delta / T))
                    chance = random.random()

                    if chance < probability:

                        accept = 1

                    else:

                        accept = 0

                # accept
                if accept == 1:

                    S = newS

                    violationsS = violationsNewS

                    costS = costNewS

                    # If the new solution is feasible
                    if violationsS == 0:

                        # The new best feasible will be the minimum between the new solution and the best feasible
                        nbf = min(costS, bestFeasible)

                    else:

                        nbi = min(costS, bestInfeasible)

                    # If the new solution is the best feasible or the best infeasible
                    if nbf < bestFeasible or nbi < bestInfeasible:

                        bestTime = time.process_time()
                        print("Cost = " + str(nbf) + " at " + str(bestTime) + " seconds")  # Print the cost and the time

                        # Reset variables to zero.
                        reheat = 0
                        counter = 0
                        phase = 0
```

```python

                        bestTemperature = T

                        # Update best feasible and infeasible costs
                        bestFeasible = nbf
                        bestInfeasible = nbi

                        # Strategic oscilation:
                        if violationsS == 0:
                            weight = weight / teta
                        else:
                            weight = weight * teta
                else:

                    counter += 1

            # If counter exceeded maxC
            phase += 1
            T = T * beta

            print("Cooling " + str(phase) + " T = " + str(T) + " Cost = " + str(nbf) + " at " + str(
                bestTime) + " seconds")

        # When phase exceeds maxP, reheat the system
        reheat += 1

        T = 2 * bestTemperature

        print("Reheating")
    if bestFeasibleSolution is not None:
        return bestFeasibleSolution
    else:
        return bestInfeasibleSolution


time.process_time()
final_solution = simulated_annealing(maxP, maxC, maxR, T, beta, weight, teta)

print(final_solution)
```

8.2.4 Sensitivity Analysis (Sobol's Method)

```python
from SALib.analyze import sobol
from SALib.sample import saltelli
import numpy as np

# 1. Define the problem
problem = {
    'num_vars': 2,
    'names': ['x', 'y'],
    'bounds': [
        [0.25, 393.55],   # x range
        [98, 19596]    # y range
    ]
}

# 2. Generate samples
N = 1024
param_values = saltelli.sample(problem, N)

# 3. Evaluate the model
def model(x, y):
    return x**2 + 10*y

z = np.array([model(x, y) for x, y in param_values])

# 4. Compute Sobol indices
Si = sobol.analyze(problem, z)

# 5. Print results
print("First-order Sobol indices:")
print(f"S_x = {Si['S1'][0]:.4f}")
print(f"S_y = {Si['S1'][1]:.4f}")
```